Ensemble-Based Multi-Disease Prediction Using Machine Learning Models

A major project report submitted in partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering / Information Technology

Submitted by

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May 2025

SUPERVISOR'S CERTIFICATE

This is to certify that the major project report entitled **'Ensemble-Based Multi-Disease Prediction Using Machine Learning Models'**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is a bonafide project work carried out under my supervision during the period from July, 2024 to May, 2025.

I have personally supervised the research work and confirm that it meets the standards required for submission. The project work has been conducted in accordance with ethical guidelines, and the matter embodied in the report has not been submitted elsewhere for the award of any other degree or diploma.

Date: 10th May, 2025

Place: JUIT

Supervisor's name: Dr. Meghna Dhalaria and Dr. Ravi Sharma

Designation: Assistant Professor
Department: CSE & IT

CANDIDATE'S DECLARATION

We hereby declare that the work presented in this report entitled **'Ensemble-Based Multi-Disease Prediction Using Machine Learning Models'** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering / Information Technology** submitted in the Department of Computer Science & Engineering and Information Technology**,** Jaypee University of Information Technology, Waknaghat is an authentic record of my own work carried out over a period from July 2024 to May 2025 under the supervision of **Dr. Meghna Dhalarai and Dr. Ravi Sharma .** The matter embodied in the report has not been submitted for the award of any other degree or diploma.

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Supervisor Name: Dr. Meghna Dhalarai and Dr. Ravi Sharma

Designation: Assistant Professor

Department: CSE & IT

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ABSTRACT

The increasing prevalence of chronic diseases such as heart disease, diabetes, and Parkinson's disease necessitates innovative approaches for early diagnosis and efficient management. This project, titled *"Ensemble-Based Multi-Disease Prediction Using Machine Learning Models,"* leverages advanced ensemble learning techniques to predict the likelihood of these diseases based on patient data. Ensemble learning, which combines multiple machine learning models to enhance predictive accuracy and robustness, serves as the foundation of this study.

The primary objective of this project is to develop a scalable and accurate prediction system capable of analyzing diverse datasets for multi-disease diagnosis. Data preprocessing techniques such as normalization, feature selection, and handling of missing values are employed to prepare the datasets effectively. The ensemble models used include Random-Forest, Gradient Boosting, and XGBoost, among others, which are evaluated for performance using metrics like accuracy, precision, recall, F1-score, and ROC-AUC. Cross-validation ensures the reliability and generalizability of the predictions.

This project explores the unique challenges of creating a unified framework for multi-disease prediction. Heart disease, diabetes, and Parkinson's disease were chosen for their global health impact and the availability of open-source datasets. By integrating disease-specific features into a combined prediction model, the system achieves competitive accuracy compared to standalone models, while providing insights into feature importance for interpretability.

The results demonstrate that the ensemble-based approach outperforms individual models in terms of accuracy and robustness, achieving an overall improvement in prediction efficiency. This reinforces the potential of machine learning as a tool for healthcare innovation.

Future work includes expanding the framework to cover additional diseases, incorporating deep learning models for more complex datasets, and deploying the system as a user-friendly application for clinical use. By addressing key gaps in multi-disease prediction, this project contributes to the broader goal of accessible and reliable healthcare solution

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Chronic illnesses such as heart disease, diabetes, Alzheimer's and Parkinson's disease are the most dramatic illnesses of our time. They account for a huge percentage of death and chronic disability globally. They not only have their bodily impact, they also introduce families emotional suffering and enormous economic load to health care institutions and economies. Control of such diseases is mostly based on early detection and early treatment. Most chronic illnesses would be treated, or even prevented, if they had been detected in the early stage, with far better results to the patients and lower healthcare costs. Traditional ways of diagnosis are time-consuming, expensive, and expert-requiring, and are thus less accessible, particularly to low-resource settings [1]. These obstacles are to be surpassed by new technologies that are cost-effective and easy to operate.

New advancements in machine learning, an offshoot of artificial intelligence, offer promising answers to such questions.Machine learning is able to process and manage larger volumes of data, detect patterns difficult for the human eye to identify, and make extremely accurate predictions. Such abilities are being utilized in this project, and using specifically ensemble learning techniques to predict the probability of heart disease, diabetes, Alzheimer's and Parkinson's disease. Ensemble learning takes the best of several predictive models to build accuracy, reduce error, and maximize reliability [2]. It is especially well-suited to medical applications where reliability and accuracy are most essential.

The overall objective of this project is to create a system that will be able to analyze patient data and forecast the risk of such diseases with high accuracy. By combining data from various sources and using advanced machine learning algorithms, the system should yield a scalable, efficient, and user-friendly diagnostic tool. The tool can potentially fill vast healthcare delivery gaps through the offer of low-cost and highly accessible early detection features.

Also, this project demonstrates the ability of artificial intelligence (AI) to revolutionize the healthcare industry. As healthcare systems globally are under increasing demands and resources are dwindling, the advent of AI brings with it a chance to fill gaps and become more efficient. This project, through the application of machine learning algorithms, demonstrates how sophisticated technology can be applied to tangible, life-saving interventions in the medical field. By providing healthcare professionals with accurate, data-driven information, the project suggests an early intervention strategy towards diagnosis and treatment. Not only does this enhance the quality of care but also eliminate diagnostic delays, enabling interventions to be performed earlier, with the potential to alter the course of disease. Also, by overcoming the shortcomings of traditional diagnostic methods, this system has been developed to reduce the socioeconomic impact of chronic disease, providing timely and accurate care to high-risk populations.

At a general level, the project acknowledges the necessity of using AI-based solutions to address world healthcare issues in an efficient manner. It is interrelated and multi-faceted in nature and demands a multi-faceted approach of treatment in cases of chronic diseases such as heart disease, diabetes, Alzheimer's and Parkinson's [3].AI possesses the unique ability of processing large volumes of data, detecting complex patterns, and delivering actionable forecasts that can impact decision-making and is thus an invaluable asset in modern medicine. It aims to become part of a future where technology and medicine complement each other in lowering mortality rates, improving patient outcomes, and enhancing the quality of life in populations. Lastly, the project is nearer to introducing advanced AI techniques to everyday healthcare practice, providing scalable, affordable, and effective solutions. Through its emphasis on early prediction and integrated disease management, it addresses the urgent need for innovative solutions for chronic disease control. Besides personal benefit, it can also revolutionize public health strategies, minimize healthcare inequalities, and lead the way towards a more equitable and resilient healthcare system worldwide.

1.2 PROBLEM STATEMENT

Early and correct diagnosis of chronic diseases remains a massive healthcare challenge in the contemporary world. The chronic diseases such as heart disease, diabetes, and Parkinson's disease have an insidious onset period, and their symptoms appear later [4]. Delayed diagnosis diminishes the efficacy of the treatment and may result in serious complications, decreased quality of life, and even death [4]. Early diagnosis is required to control the diseases, but the existing diagnostic methods based on conventional methods are not effective. These are based on costly tests, sophisticated equipment, and skilled personnel, which are not only resource-intensive but also not available, particularly in low-resource or resource-limited settings. In spite of the advancement in technology, there remains a gap in the availability of low-cost, effective, and scalable diagnostic devices.

Although machine learning offers a potential solution to disease prognosis, most current models are narrow in focus. Most predict one disease and do nothing to address the wider interconnected scope of chronic diseases. Diabetic patients, for example, are usually also at risk for heart disease, but systems today are seldom configured to detect overlaps of this type. Furthermore, most machine learning models are also plagued by low accuracy due to inconsistent data, inadequate feature selection, or ineffective algorithmic performance. The problem of this project is to address these shortcomings by creating an ensemble-based machine learning model with the capacity to predict the probability for a cluster of diseases—heart disease, diabetes, Alzheimer's and Parkinson's disease—within a single system. Ensemble learning, wherein strengths of numerous algorithms are combined, is a robust way of increasing accuracy, reliability, and generalizability.

By integrating data from various datasets and taking advantage of the complementarity of strengths among models such as Random Forest, Gradient Boosting, and XGBoost, the system attempts to construct a more complete solution.
The long-term goal of this project is to create a diagnostic system that not only increases the accuracy of the predictions but is also cost-effective and scalable. Such a system has the potential to revolutionize healthcare by enabling early diagnosis,

preventing treatment delays, and saving lives in the long term. Through this project, we hope to close some of the most critical gaps in existing diagnostic tools and enable the development of technology-based solutions for healthcare.

1.3 OBJECTIVES

The overall goal of this project is to create an ensemble-based machine learning model that can effectively forecast the probability of four chronic illnesses: heart disease, diabetes, Alzheimer's and Parkinson's disease. In pursuit of this major objective, multiple specific objectives have been established that cumulatively lead to developing a strong, scalable and robust solution.

1.3.1 DATA COLLECTION AND PREPROCESSING

The basis of any machine learning model is the quality of data on which it's trained. The aim of this project is to gather real and heterogeneous datasets of heart disease, diabetes, Alzheimer's and Parkinson's disease from publicly available sources and ensure that the data sample has heterogeneous population and range of medical conditions. Preprocessing shall involve data cleansing, managing missing values , normalization of features, and identification of significant variables influencing prediction of the disease. Preprocessing makes data provided to the model accurate, consistent, and valid.

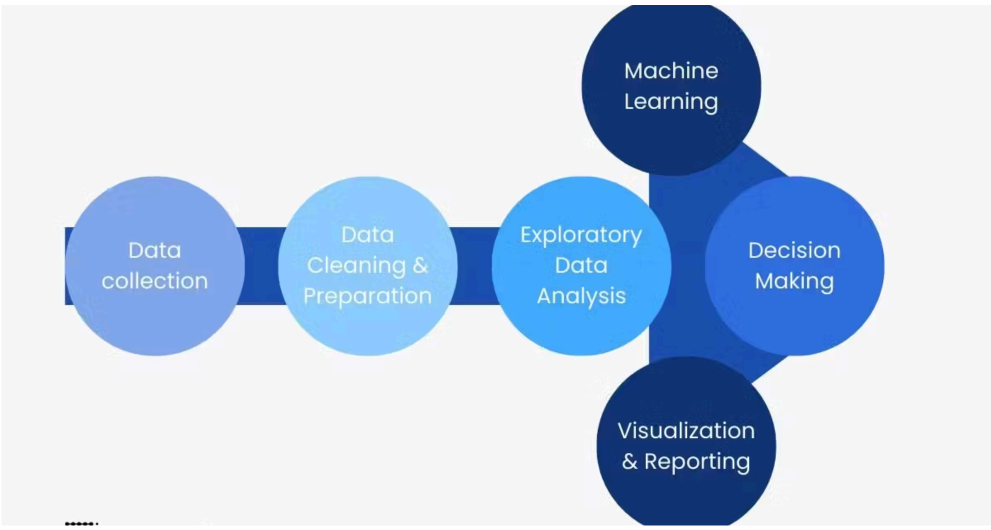

Fig 1.1 *Data Processing Steps*

1.3.2 MODEL DEVELOPMENT USING ENSEMBLE TECHNIQUES

Ensemble learning is a method in which various models are combined to improve the prediction accuracy. The focus of this project is on implementing ensemble techniques such as Random-Forest, Gradient Boosting, and XGBoost. These methods leverage the power of different algorithms and therefore guarantee that the system is not vulnerable to mistakes, hence boosting robustness.In this project, the aim is to create models that are able to process huge datasets with efficiency, find patterns, and make very accurate predictions.

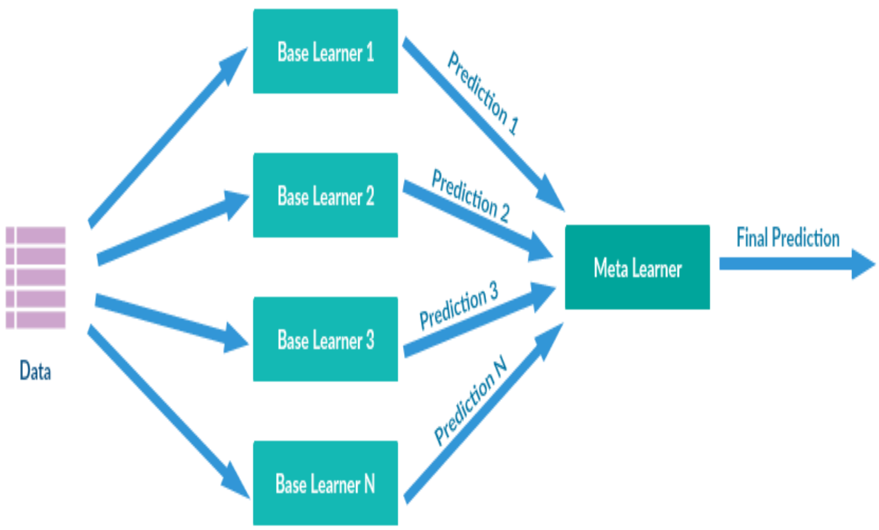

Fig 1.2 *Ensemble Learning*

1.3.3 PERFORMANCE EVALUATION AND OPTIMIZATION

One of the primary objectives is to test the models developed using common performance measures such as accuracy, precision, recall, F1-score, and ROC-AUC. These models will be tuned and optimized based on these tests so that they become as efficient and reliable as possible. These models must be standardized to be implemented in various datasets so that they can be used practically in actual applications.

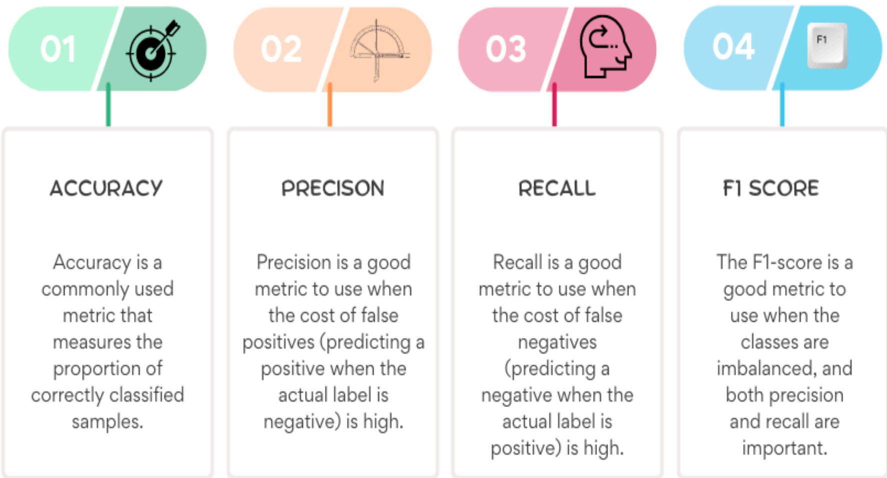

Fig 1.3 *Evaluation Metrics*

1.3.4. MULTI-DISEASE PREDICTION FRAMEWORK

Unlike the majority of existing solutions that address one disease at a time, this project will create a system that predicts heart disease, diabetes, Alzheimer's and Parkinson's disease. This multi-disease approach is designed to address the interconnectedness of the long-term diseases and improve diagnostic efficiency.For consolidating the predictions in a platform, the system can offer medical practitioners a better diagnostic tool.

1.3.5. SCALABILITY AND USABILITY

Yet another main goal is to make the system scalable and user-friendly. The system to be developed should be scalable to new data sets and scalable to other diseases in the future. The interface or platform should be user-friendly as well, making it easy for healthcare providers and researchers to use with minimal training.

1.3.6. INTERPRETABILITY AND TRANSPARENCY

Creating trust in machine learning models is a matter of making them explainable. This project is centered on discovering and documenting the most significant features that are accountable for making the predictions, i.e., certain biomarkers or lifestyle factors. Providing clear insight into the way the model is making its predictions guarantees transparency and builds trust in its use in healthcare.

Overall, the objective of the project is to develop a resilient, efficient, and scalable multi-disease prediction model. Using the latest machine learning methods, the project aims to fill gaps in existing diagnosis protocols and provide improved health results.

1.4 SIGNIFICANCE AND MOTIVATION OF THE PROJECT WORK

The motivation behind this project is the pressing need for better diagnostic technologies in medicine, especially for chronic diseases like heart disease, diabetes, and Parkinson's disease. These diseases are an epidemic globally, affecting millions of individuals each year. Late diagnosis postpones their impact, as early medical intervention can retard disease progression, enhance patient outcomes, and greatly lower healthcare system costs.Although tremendous advances have been made in the medical science and technology community, the reality is that access to early and precise diagnostic machinery is unequal. This is particularly true in resource-constrained environments, where few and inexpensive and scalable solutions are still not able to reach many of the needy. The worth of this project is the potential to transform the diagnosis of chronic disease.

Taking advantage of the strength of machine learning, or more specifically ensemble methods, this project seeks to create a comprehensive system to predict a variety of diseases accurately [6]. Ensemble learning, through the ensemble predictive power of multiple algorithms, is particularly suited to healthcare, where accuracy and reliability are the utmost importanceThis approach considers not only the inherent complexity of each disease but also the interconnected nature of the diseases themselves, in the knowledge that many share common risk factors and frequently present in the same patients. The second driving force is the potential of this project to fill compelling gaps in the delivery of healthcare.

Modern diagnosis tends to be labor-intensive, costly, and dependent on experts.Parkinson's disease, for instance, usually mandates advanced imaging and neurological evaluation, and heart disease and diabetes are based on a cascade of expensive laboratory testsThese pose barriers for residents in low-income districts or rural villages with limited health care infrastructure [7]. In leveraging machine learning to construct an evidence-based, computer-based platform for diagnosis, this project has the solution that is efficient and scalable. As valuable as its immediate applied benefits, this project also has an expanded vision: using artificial intelligence to push the boundaries of precision medicine. Machine learning is better suited to manage big, complex data sets and extract subtle patterns beyond the eye of human beings. Translating these strengths into predicting chronic disease not only enhances diagnostic quality but also makes actionable information more available to health care providers [8].For example, primary risk factor and predictor biomarker identification can be used to allow individualized prevention and treatment options for patients. The project is also motivated by its potential to impact public health.

Non-communicable disease causes a large proportion of worldwide death and disability. By allowing early diagnosis through a provided tool, this project can be expected to make an impact on complication prevention and enhanced long-term health outcomes [9]. Furthermore, it is responding to the call for healthcare solutions that are equitable and that are able to reach diverse socioeconomic and geographic populations. This project is a stepping stone to that vision, not only emphasizing technological innovation but also its ethical and practical use for the greater good. In short, the worth of this project is that it has the potential to merge innovative technology with actual health care requirements. Encouraged by the limitation of identifying chronic diseases and the potential to make an impact, this study seeks to create an affordable, efficient, and reliable diagnosis tool. In exchange, it hopes to reduce the burden of chronic diseases, enhance patient care, and enable cost-effective delivery of health care across the world.

1.5 ORGANIZATION OF PROJECT REPORT

This report is organized into six chapters to give an overall idea of the project. The project is presented in Chapter 1, including background, problem statement, objectives, and motivation. Chapter 2 provides a literature survey of research on machine learning in the healthcare field and the gaps in research that this project fills.

Chapter 3 is devoted to the development of the system, data collection, preprocessing, model implementation, and architecture design. Chapter 4 addresses the testing strategy, test cases, evaluation metrics, and results. Chapter 5 offers project results, including an in-depth analysis and comparison with other solutions.

Finally, Chapter 6 provides an overview of findings, contributions to the field, limitations, and future work recommendations for concluding the report. Each chapter is designed to provide a clear and concise description of the project so that readers can build a complete understanding of its aims, method, and impact.

CHAPTER 2: LITERATURE SURVEY

2.1 OVERVIEW OF RELEVANT LITERATURE

S No.	Title	Authors	Year	Techniques Used	Results	Conclusion
1.	Heart disease prediction using machine learning algorithms	Harshit Jindal, Sarthak Agrawal	2020	logistic regression and k-nearest neighbor	KNN algorithm, 88.52% accuracy.	This study emphasizes the role of machine learning algorithms in the effective and efficient early detection and prevention of heart disease.
2.	Heart disease prediction using machine learning techniques	V.V. Ramalingam, Ayantan Dandapath, M Karthik Raja	2018	Decision Trees,Naive Bayes,K–Nearest Neighbors, Support Vector Machines (SVM),Artificial Neural Networks	KNN algorithm, 89.12% accuracy.	The findings of this investigation come to support the idea that with this technology, heart diseases can be predicted with a degree of cons.
3.	A Study on Heart Disease Prediction Using Machine Learning Algorithms	Rai, S., Durgude, U., Dhavan, A., Singh, P., & Ojha S.	2024	Decision Trees, Support Vector Machines (SVM), Naïve Bayes, and K-Nearest Neighbors (KNN)	SVM algorithm, 90.32% accuracy.	The work concludes that machine learning algorithms can go a long way in the early diagnosis of heart disease. Through proper choice

						of the algorithms and fine tuning of the parameters, healthcare experts can obtain meaningful predictive models that can be used for early recognition of at-risk patients and subsequent early management.
4.	Heart Disease Detection by Using Machine Learning Algorithms and a Real-Time Cardiovascular Health Monitoring System	Nashif, S., Raihan, Md. R., Islam, Md. R., & Imam, M.H.	2018	Support Vector Machines (SVM), Naïve Bayes, and K-Nearest Neighbors (KNN)	Support Vector Machine, accuracy 97.53%, sensitivity 97.50%, and specificity 94.94%.	Real time monitoring is one of the perfect applications of the IoT-ML combination that provides early diagnosis and monitoring of heart disease. This approach not only helps to increase the accuracy of diagnosing diseases but also helps to monitor patients' conditions constantly, which could have a positive impact on decreasing mortality connected with cardiovascular

						diseases.
5.	Prediction of Diabetes Disease Using Machine Learning Model	A. Sharma, K. Guleria, and N. Goyal	2021	Decision Trees, Naive Bayes, Artificial Neural Network Classifiers,Logistic Regression.	The Decision Tree classifier achieved accuracy 90.89%.	The research paper presents a broad framework for predicting Diabetes mellitus with the help of supervised Machine learning algorithms well. This accounts for the fact that the study attaches significant value to early identification in moderating the various severe health risks associated with diabetes.
6.	Diabetes Prediction Model Using Machine Learning Techniques	S. K. S. Modak and V. K. Jha	2023	Logistic Regression, Support Vector Machine (SVM), Naive Bayes, and Random-Forest. Also, It uses other ensemble learning algorithms such as XGBoost, LightGBM, Cat Boost, Ada Boost & Bagging.	The Decision Tree algorithm achieved 90.32% accuracy.	The paper claims that using more complex ensemble learning techniques, such as CatBoost, more efficiency and effectiveness can be gained for constructing diagnostic indicators for diseases such as diabetes.

7.	Improving Healthcare Prediction of Diabetic Patients Using KNN Imputed Features and Tri-Ensemble Model	K.Alnowaiser	2024	Random-Forest, Extreme Gradient Boosting, and Extra Trees Classifier.	KNN with an accuracy of 97.49%, precision of 98.16%, recall of 99.35%, and F1 score of 98.84% percent.	KNN imputation for missing data integrated with the tri ensemble voting classifier contributed highly to improving the prediction of diabetes. This might promise a tool which can provide early diagnosis and intervention for patients by health professionals.
8.	A Diabetic Disease Prediction Model Based on Classification Algorithms	R. Ahuja, S. C. Sharma, and M. Ali	2019	Support Vector Machine, Multi-Layer Perceptron, Logistic Regression, Random-Forest, and Decision Tree.k-fold cross-validation.	MLP had an accuracy of 78.7%, a recall of 61.26%, a precision of 72.45%, and an F1 Score of 65.97% for k = 4.	The study has demonstrated that out of all the algorithms evaluated, the MLP classifier performed superior for the prediction of diabetes and could clinically be beneficial to a certain extent for its early diagnosis and intervention.
9.	Cardiovascular Diseases Prediction by Machine Learning Incorporation with Deep Learning	Mohammad A. Al-Dhif,Baraa G. Jamil,M. Sheraz Ahmed,Kamran Iqbal,Muhammad	2023	A hybrid model will be developed that integrates the selected features into deep neural networks.	The hybrid model performed with increased predictive accuracy against a regular ML model.	Combining Deep-Learning techniques with algorithms of machine learning develops an advanced predictive

		Naeem,Shahid Mumtaz				capability concerning diseases.
10.	Effective Heart Diseases Predictions Using Machine-Learning Techniques	Chintan M. Bhatt,Parth Patel,Tarang Ghetia, Pier Luigi Mazzeo	2024	Random-Forest, Decision Tree, Multilayer Perceptron, and XGBoost	XGBoost model returns the best score among the models applied-most effective for heart disease predictions.	The integration of k-modes clustering with Huang's initialization and enhanced ML algorithms such as XGBoost enhances performance significantly in the prediction capability related to cardiovascular disease for supporting needed clinical treatment.
11.	Prediction of Parkinson's disease using a hybrid XGBoost and ensemble learning with voting classifier	Jai Narayan Verma, R. Anandan, Shubham Jain, Ritu Vijay, Rajendra Kumar Malviya	2023	XGBoost,Random-Forest (RF),Decision Tree (DT),Support Vector Machine (SVM),K-Nearest Neighbors (KNN)	XGBoost with ensemble voting, and it has 95.57% accuracy.	With the integration of ensemble learning methodology and utilization of the strength of XGBoost, the research improves the prediction power for Parkinson's disease.
12.	Predicting patients with Parkinson's disease using Machine Learning and ensemble voting technique	Shawki Saleh, Bouchaib Cherradi, Oussama El Gannour, Soufiane Hamida, Omar	2023	KNN , SVM,Random-Forest (RF),Multilayer Perceptron (MLP),XGBoost.	Voting Classifier 1 (Dataset I): Accuracy: 96.41% F1-Score: 97.59% Voting Classifier 2	Applying machine learning with optimally calibrated parameters and ensemble methods considerably

		Bouattane			(Dataset II): Accuracy: 97.35% F1-Score: 98.23%	enhances Parkinson's disease prediction.
13.	Multistage Ensemble-Learning-Based System for Early Prediction of Parkinson's Disease Using Voice Feature Set	Riya Gupta,Meenu Gupta,Monica Sharma,Harsimrat Kaur	2023	Random-Forest,Support Vector Machine (SVM),Gradient Boosting, Voting Classifier	End model accuracy: 98.79% F1-Score: 0.9879	The present research indicates that by combining various machine learning methods in stages, it is possible to significantly enhance Parkinson's disease prediction using voice data.
14.	Ensemble Approach for Early Prediction of Parkinson's Disease Using Voice Features	Simran Jeet,Abhay Rawat,Abhishek Tiwari,Shubham Sharma,Ramesh Kumar Sunkaria	2021	Decision Tree,Random-Forest,Gradient Boosting,XGBoost. Created a composite model by using the best (Random-Forest, Gradient Boosting, XGBoost)	The ensemble model outperformed individual models in terms of accuracy. Best model accuracy: 94.87%	This research confirms that with an ensemble methodology to merge multiple machine learning models, the prediction of Parkinson's disease is significantly improved.
15.	Detection of Alzheimer's Disease Using Machine Learning Models	Subramanian Palaniappan ,Mohamed Saeed Almalki, Akmal S. Malik,	2022	Decision Tree,Random-Forest,Gradient Boosting,XGBoost. Created an ensemble	Ensemble model was more accurate and consistent compared to	This study establishes that using ensemble learning to combine multiple machine learning

		Naveed Khan, Mohamed Elhoseny, Ahmad Taher Azar, M. Shamim Hossain, Abdullah Alharbi		model by merging top performers (RF, GB and XGBoost) to enhance accuracy and strength.	individual models.Best model accuracy: 94.87%.	models considerably enhances prediction of Alzheimers's disease.
16.	An ensemble learning-based framework for detection using hybrid feature space	K. Murugesan, M. A. Jabbar	2022	Decision Tree (DT),Random-Forest (RF),Gradient Boosting (GB),XGBoost, SVM, KNN An ensemble model was constructed using stacking.	Ensemble model accuracy: 94.87%. Outperformed individual models in all major metrics.	The research verifies that ensemble learning significantly enhances Alzheimer's disease predictive accuracy.
17.	A Multi-Sensor Fusion Approach for the Detection of Alzheimer’s Disease Using ML Techniques	Taner Arslan, Ahmet Sayar	2024	Support Vector Machine (SVM),Random-Forest,K-Nearest Neighbors (KNN),Gradient Boosting.	The ensemble model that used all four modalities performed better models.Best accuracy: 99.33%.	Fusing multiple behavioral data sources with ensemble ML techniques greatly enhances Alzheimer's disease prediction.
18.	A Hybrid Deep Learning–Acoustic Features Framework for Detecting Alzheimer's Disease Using Sustained Phonation	Rohit Arora, Shubham Neupane, Christopher Leung, Antonio Jimeno Yepes, Rajib Rana	2024	SVM, KNN, Random-Forest, Gradient Boosting.End model: CNN-acoustic model hybrid fusion for better performance and generalizatio	The hybrid model considerably outperformed the models that employed only acoustic or deep features in	Blending domain-specific acoustic features with deep learning models yields more accurate and reliable detection of disease.

				n.	isolation.	
19.	Diagnosis of Alzheimer’s Disease Using Ensemble Learning Approach	Masrur Ahmed, Shahed Mahmud, Md. Mohaiminul Islam	2022	Support Vector Machine,Random-Forest, DecisionTree,K-Nearest Neighbor,XGBoost.An ensemble model of Random-Forest, XGBoost.	Maximum model accuracy: 94.87%,Ensemble model performed better than independent classifiers.	The approach is effective, non-invasive, and appropriate for supporting early diagnosis in clinics.
20.	Machine learning and deep learning for data and imaging in Alzheimer’s disease	Rania Khachnaoui, Moulay A. Akhloufi, and Petr Musilek	2022	Decision Tree, RandomForest,SVM,Deep Learning: CNN (for imaging), MLP (for structured clinical data)	Highest performance was achieved with hybrid deep learning approaches (CNN + clinical data MLP).Reported over 90% accuracy.	Ensembling clinical and imaging data using ensemble and AI models provides outcomes for early diagnosis of Alzheimer's.The method is a strong, scalable, and precise approach to decision support systems in neurology.

CHAPTER 3: SYSTEM DEVELOPMENT

3.1 REQUIREMENTS AND ANALYSIS

Building a strong multi-disease prediction system needs well-chosen datasets, effective hardware, trustworthy software, and proper pre-processing methods. For this project, we obtained datasets mostly from public sources like the UCI Machine Learning Repository, which provides extensive datasets for heart disease, diabetes, Parkinson's and Alzheimer's. These datasets contain a wide collection of features ranging from patient demographics to clinical biomarkers to test results that are needed to build a trustable prediction model.

Hardware configuration included a machine with a minimum of 8GB RAM, a multi-core processor, and an optional GPU to support computationally expensive tasks such as model training and hyperparameter tuning. Software stack included Python as the programming language at the highest level, Pandas and NumPy for data manipulation, Scikit-learn and XGBoost for deployment of machine learning models, and Matplotlib for plots. Jupyter Notebook was also employed for an interactive development environment.

Pre-processing techniques involved data cleaning to remove inconsistencies, imputation for missing value handling, and feature standardization to have consistent output for variables. Feature importance and pattern discovery in the data was the area of concern for the data analysis process, resulting in model designing.

3.2 PROJECT DESIGN AND ARCHITECTURE

The system design has been developed to tackle the complexity of multi-disease prediction effectively. The system architecture is modular with different modules for prediction, model training, and data processing. The architecture diagram has the following crucial steps:

1. Data Ingestion: Input from different datasets is put together into one format.
2. Data Preprocessing: Normalization, cleaning, and feature engineering ready the data for modeling.

3. Model Training: Ensemble learning models like Random-Forest and XGBoost are used.
4. Prediction and Evaluation: The models are trained to predict diseases, and these predictions are verified against testing data.

Work-flow diagrams also illustrate the sequential workflow of the tasks, ranging from raw data loading to generating actionable predictions. This structured architecture ensures scalability and enables straightforward integration into healthcare systems.

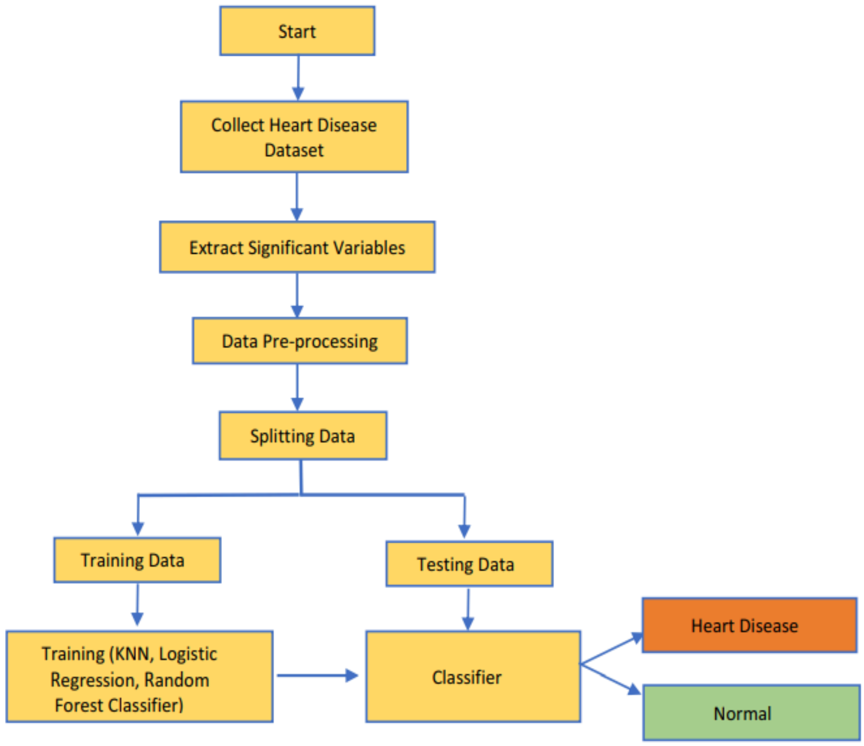

Fig 3.1 *Project Design 1*

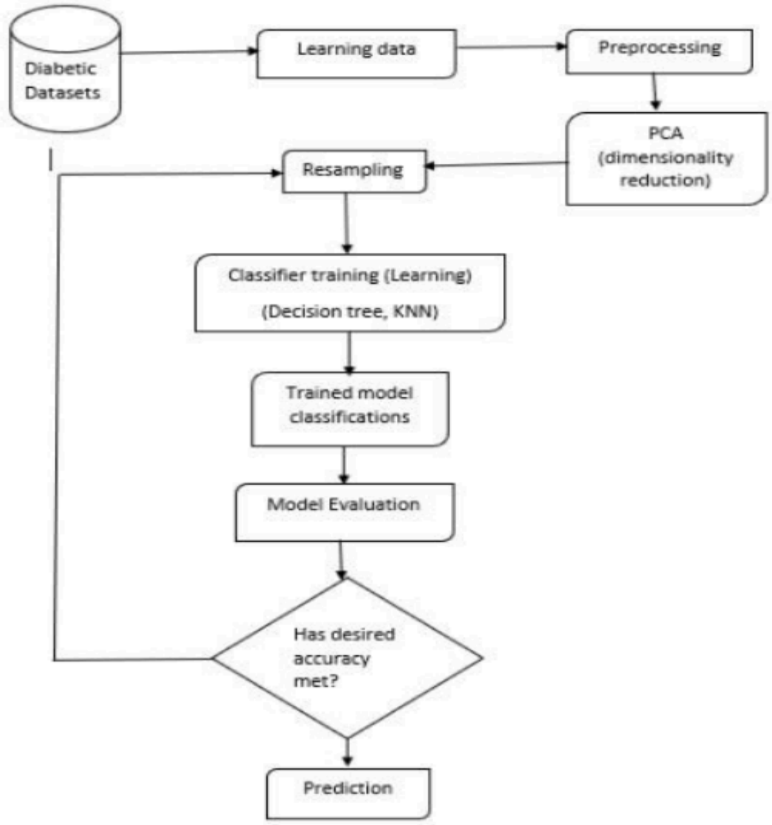

Fig 3.2 *Project Design 2*

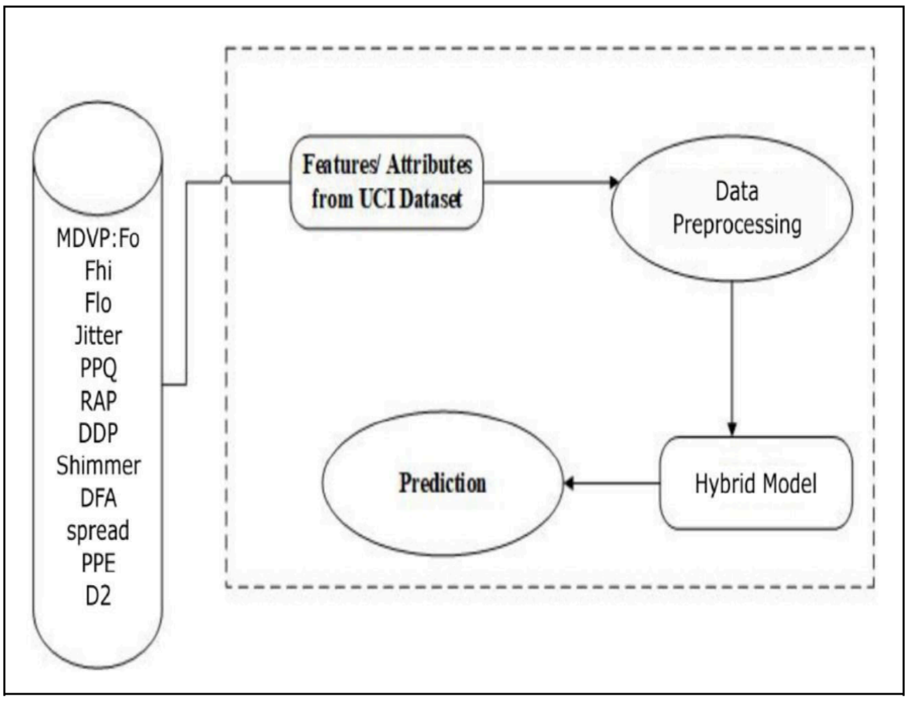

Fig 3.3 *Project Design 3*

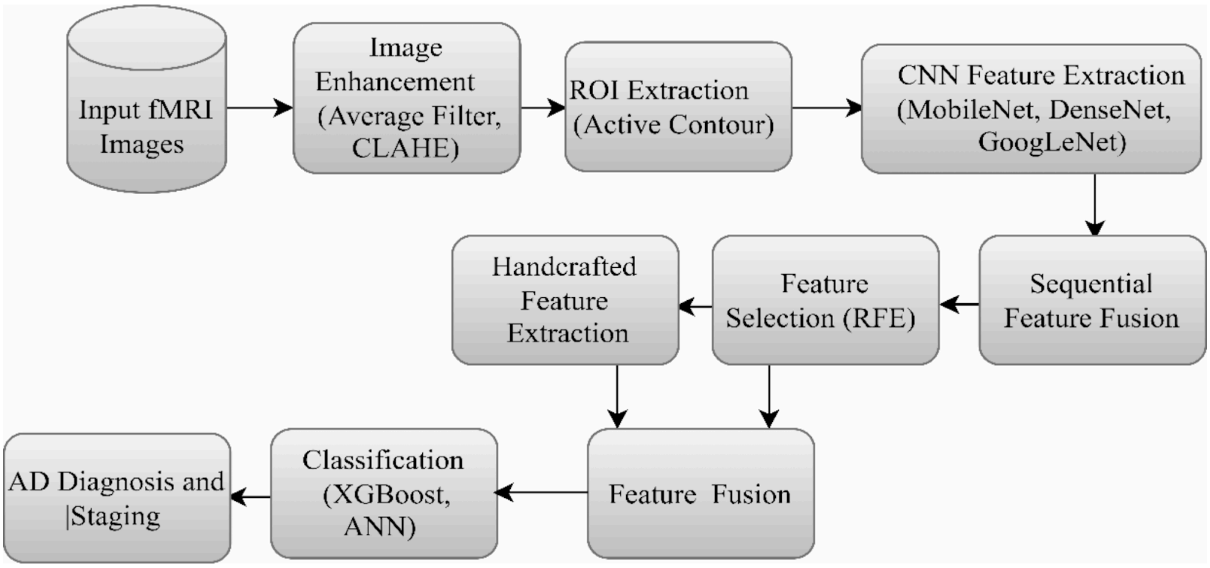

Fig 3.4 *Project Design 4*

3.3 DATA PREPARATION

Data preparation was of utmost importance for the success of this project. Missing values within the data were handled using median or mean imputation for numerical features and mode imputation for categorical features. Outliers were identified and handled using statistical methods like the interquartile range (IQR). Feature scaling techniques like Min-Max scaling were utilized to normalize numerical features.

Feature engineering was used to create derived variables for imparting predictive power. For example, body mass index (BMI) was computed from weight and height data, and age ranges were binned for readability reasons. Class imbalance, particularly in datasets with uneven distributions (e.g., Parkinson's disease datasets with few positive cases), was handled using oversampling techniques like SMOTE (Synthetic Minority Oversampling Technique).

3.3.1 HEART DISEASE DATASET

The heart disease dataset consists of 303 records, with 11 features and a binary target variable. The features include six nominal variables and five numerical variables. Notable features are age, chest pain type, cholesterol levels, and maximum heart rate achieved. The target variable indicates whether a patient is at risk of heart disease (1) or normal (0).

Before modeling, the dataset underwent thorough cleaning and preparation. Missing values, if any, were treated using imputation techniques; numerical features like blood pressure and cholesterol were filled with their median values, while categorical features like chest pain type were assigned the mode. Outliers in numerical variables such as cholesterol and oldpeak were handled using the interquartile range (IQR) method to improve data integrity.

Normalization techniques such as Min-Max scaling were applied to features like cholesterol and maximum heart rate to standardize the range, enabling models to converge faster during training. Nominal variables like chest pain type and ST slope were one-hot encoded to ensure compatibility with machine learning algorithms.

Class imbalance, though minimal in this dataset, was checked to ensure that disease cases (target = 1) were sufficiently represented. The final dataset was split into training and test subsets using an 80:20 ratio.

Features are as follows:

1. Age: Patient's age, expressed in years (numerical value).
2. Sex: Patient's gender, represented as a binary variable (Male = 1, Female = 0) (categorical).
3. Chest Pain Category: The type of chest pain the patient experienced, classified into four categories: 1 (typical angina), 2 (atypical angina), 3 (non-anginal pain), and 4 (asymptomatic) (categorical).
4. Resting Systolic Blood Pressure: Blood pressure level measured in millimeters of mercury (mmHg) while the patient is at rest (numerical value).
5. Serum Cholesterol: Cholesterol level in the patient's blood, measured in milligrams per deciliter (mg/dl) (numerical value).
6. Fasting Blood Glucose: Indicates whether the patient's fasting blood sugar level was greater than 120 mg/dl (1 = True, 0 = False) (binary categorical).

7. Resting Electrocardiogram Results: Findings from the patient's electrocardiogram taken at rest, categorized into three distinct values: 0 (Normal), 1 (ST-T wave abnormality), and 2 (Left ventricular hypertrophy) (categorical).
8. Maximum Heart Rate Achieved: The highest heart rate reached by the patient (numerical value).
9. Exercise-Induced Angina: Indicates whether the patient experienced angina as a result of exercise (0 = No, 1 = Yes) (binary categorical).
10. ST Depression Induced by Exercise: The amount of ST segment depression observed during peak exercise relative to the resting state (numerical value).
11. Slope of the ST Segment During Exercise: The direction of the ST segment during peak exercise, categorized as: 0 (Normal), 1 (Upsloping), 2 (Flat), and 3 (Downsloping) (categorical).

Target variable

1. Target: Which we will predict 1 is patient suffering from heart disease and 0 means patient is fine.

3.3.2 DIABETES DATASET

The diabetes dataset contains 768 instances, each representing a female patient of Pima Indian heritage. The dataset comprises 8 features and one binary target variable, where 1 indicates diabetes-positive and0 indicates diabetes-negative. Key features include glucose levels, blood pressure, BMI, and age.

This dataset posed challenges such as missing attribute values for features like skin thickness and insulin levels. Missing values were imputed using mean or median values based on feature distribution to preserve dataset consistency. Outliers in continuous variables like insulin were detected using box plots and corrected to improve the model’s performance.

Normalization was applied to numerical features such as glucose and BMI to scale them between 0 and 1. Feature engineering included deriving interaction terms, such as BMI multiplied by diabetes pedigree function, to uncover hidden patterns.

Class imbalance was addressed, ensuring equal representation of diabetes-positive and diabetes-negative samples in the training set. The final preprocessed dataset was split into training (80%) and testing (20%) subsets

The detailed description of all the features are as follows:

1. Pregnancy Count: The total number of pregnancies the patient has had.
2. Post-OGTT Plasma Glucose: The concentration of glucose in the patient's plasma, measured two hours following an oral glucose tolerance test.
3. Diastolic Pressure: The diastolic measurement of the patient's blood pressure, recorded in millimeters of mercury (mm Hg).
4. Triceps Skinfold Measurement: The thickness of the skin fold on the triceps, expressed in millimeters (mm).
5. Two-Hour Serum Insulin Level: The insulin concentration in the patient's blood serum, quantified in micro international units per milliliter (mu U/ml) after a two-hour period.
6. Body Mass Index (BMI): An index calculated from the patient's weight and height, specifically weight in kilograms divided by the square of the height in meters.
7. Diabetes Genetic Risk Score: A score reflecting the genetic predisposition to diabetes based on family history.
8. Patient Age: The age of the patient, expressed in years.
9. Diabetes Diagnosis: A categorical variable indicating the presence (1) or absence (0) of a diabetes diagnosis.

3.3.3 PARKINSON’S DISEASE DATASET

The collection includes 195 instances, each of which refers to the voice recording of a subject. Twenty-three features are included in the dataset, with a binary target variable, coded as 1 for a subject with Parkinson's and 0 for a healthy subject.

Some battles faced by this dataset include the need for feature scaling due to the discrepancy in the range of the numerical attributes. Outlier values were found in features such as shimmer and jitter through the use of box plots and dealt with either through data transformation or deletion for the sake of better model performance. Missing values, if any, were handled by imputing the means or medians of the feature distribution in order to keep the integrity of the dataset.

The continuous features were normalized such as fundamental frequency (MDVP:Fo) and also shimmer so that their ranges could be standardized. Feature engineering was applied to develop new features or interaction terms such as a combination of jitter and shimmer values to uncover some hidden patterns in the data.

Finally, after preprocessing, data were divided into 80%:20% for training and testing subsample respectively; an imbalanced class problem usually occurs when healthy subjects (class 0) do not have equal numbers compared to patients who have Parkinson's disease (class 1). Smoothing methods such as Synthetic Minority Oversampling Technique (SMOTE) could be applied to balance distributions of the two classes.

The detailed description of all the features are as follows:

1. MDPV:Fo(Hz): Fundamental frequency of vocal cord vibration in Hz.
2. MDVP:Fhi(Hz): Max. frequency in Hz of the vocal signal.
3. MDPV:Flo(Hz): Min. frequency in Hz of the vocal signal.

4. MDVP:Jitter(%): Percentage of jitter that measures the frequency variation in the voice.
5. MDVP:Jitter(Abs): Absolute jitter is the total variation of frequency.
6. MDPV:RAP: relative average perturbation, quantification of jitter.
7. MDPV:PPQ: another measure of jitter-the pitch perturbation quotient.
8. Jitter:DDP: Jitter computed using the spanning of differences of differences.
9. MDPV:Shimmer: Amplitude variation in the voice signal.
10. MDPV:Shimmer(dB): Shimmer expressed in decibels-amplitude variations measured.
11. Shimmer:APQ3: Amplitude Perturbation Quotient (measure of amplitude perturbation).
12. Shimmer:APQ5: Another measure of amplitude perturbation (Amplitude Perturbation Quotient);
13. MDVP:APQ: Amplitude Perturbation Quotient (mean value of amplitude perturbation).
14. Shimmer:DDA: A measure of shimmer based on differences between two consecutive signal periods.
15. NHR: Noise-to-Harmonics Ratio, encompassing noise proportion in the voice signal.
16. HNR: Harmonics-to-Noise Ratio, encompassing harmonic proportions relative to noise.
17. Status: the target variable where 1 indicates Parkinson's disease and 0 indicates a healthy individual.
18. RPDE: Recurrence Period Density Entropy; capturing irregularities in vocal patterns.
19. DFA: Detrended Fluctuation Analysis, finding fractal-like scaling properties of the signal.
20. Spread 1: First spread of the recurrence points in the signal representing the distribution of voice features.
21. Spread2: A second measure of the spread in the recurrence plot.
22. D2: Correlation dimension, a measure of the fractal structure of the signal.
23. PPE: Phase Plane Entropy, capturing the complexity of the signal.

3.3.4 ALZHEIMER’S DISEASE DATASET

This is the Alzheimer's disease dataset from Kaggle that contains over 5,000 MRI brain scan images obtained from different sources. Every scan shows a person going through one of four stages of cognitive health: Non-Demented, Very Mild Demented, Mild Demented, or Moderate Demented. These labels indicate the progress of the disease in the brain structure, which progressively differs between the diseases. The ventricle is known to grow in size as a result of atrophy, indicating the amount of black space in the brain, especially in the temporal and parietal lobes.

The MRI scans were from different sources and thus had problems like varying image-size and contrast. Uniformity was established by resizing all images to the same size followed by normalization and contrast enhancement measures. All were done to maintain image quality and structure for diagnostic value retention. Augmentations applied to the images for improving the model robustness and addressing variations in brain orientation were rotation, flipping, zooming, and shifting.

Furthermore, the class imbalance where the Non-Demented class held too many images compared to Mild and Moderate Demented categories was dealt with by oversampling the under-represented classes and applying class weights during model training to assure balanced training across all classes. The pixel intensity values were normalized from 0-1 for input into convolutional neural networks. The finalized preprocessed dataset was split into training (80%) and test (20%) subsets.

A broadly defined draft seen in relation to all the considered classes is as follows:

- No patient is considered demented unless overt signs of Alzheimer disease or evidence of cognitive impairment is found.
- Very mild deterioration: Early changes in brain structure but possibly still quite subtle signs of dementia.

- Mild decline: Involving mild alterations in areas of the brain including the parietal and temporal lobes associated with mild cognitive impairment.
- Moderate dementia: Widespread structural changes and loss of mass in regions of marked cognitive impairment.

3.4 IMPLEMENTATION

The project implementation stage involved a systematic process of building a strong multi-disease prediction model using ensemble machine learning techniques. The focus was to ensure high prediction performance without compromising on the interpretability and scalability of the model.

3.4.1 ENSEMBLE LEARNING MODELS

The system employed a collection of ensemble algorithms, which were:

- Random-Forest Classifier: It is famous for its ability to handle big data sets and minimize overfitting by bagging.
- Gradient Boosting Classifier: A sequence of ensemble techniques that tries to reduce errors by giving more importance to misclassified instances.
- XGBoost Classifier: An accelerated and improved variant of gradient boosting that also supports parallel processing and regularization.
- AdaBoost Classifier: It combines many weak learners to create a strong learner by learning from errors in training data.
- Bagging Classifier: Averages the outputs of numerous models to reduce variance.
- Extra Trees Classifier: Builds multiple trees with random splits, which increases variance reduction.
- Voting Classifier: Combines the predictions of several algorithms using majority or weighted voting.
- Stacking Classifier: Utilizes more than one algorithm and employs a meta-model to optimize overall performance.

Each algorithm was trained and evaluated on the heart disease, diabetes , parkinson’s and Alzheimer's datasets, with results compared based on metrics such as accuracy, precision, recall, and F1 score.

3.4.2 MACHINE LEARNING ALGORITHMS

- Support Vector Machine (SVM): A strong classing machine based on maximum margin, where the hyperplane creates the optimum separation of the classes. Useful in high dimension, robust against overfitting with the aid of kernel tricks.
- K-Nearest Neighbors (KNN):An algorithm that classifies based on the majority vote of its neighbors. It is simple and effective, especially for small data sets with well-separated classes.

3.4.3 DEEP LEARNING ALGORITHM

Convolutional Neural Network (CNN): It is introduced specifically for image data and is capable of automatically learning spatial hierarchies of features through convolutional layers. It captures all kinds of patterns like textures, edges, and different shapes from MRI scans that make it appropriate for the classification of various stages of Alzheimer's disease. It reduces the need for manual feature extraction and itself learns from raw pixel data. This automatically increases the model accuracy and generalizations in image-based medical diagnosis

3.4.4 HYPERPARAMETER TUNING

To optimize the models, grid search and cross-validation techniques were employed. Hyperparameters such as the number of trees, learning rate, and maximum depth were fine-tuned for each model. For example:

- Random-Forest: The number of estimators was tuned to balance between accuracy and computational cost.
- XGBoost: Parameters like learning rate, gamma, and max depth were adjusted to enhance model efficiency.

- Gradient Boosting: Learning rate and number of boosting stages were optimized for better results.

3.4.5 CODING AND TOOLS

The implementation was carried out in Python using libraries such as:

- Numpy and Pandas for data manipulation.
- Matplotlib and Seaborn for data visualization and understanding feature distributions.
- Scikit-learn for implementing machine learning models like Random-Forest, Gradient Boosting, and others.
- XGBoost for leveraging its high-performance gradient boosting algorithm.
- Convolutional Neural Network (CNN) construction and training powered by TensorFlow and Keras will be used for Alzheimer's disease classification from MRI scans.
- Keras's preprocessing utilities would be used for augmenting, normalizing, and resizing images in preparation for CNN input.

A modular coding approach ensured reusability and maintainability. Key modules included data preprocessing, feature selection, model training, hyperparameter tuning, and evaluation. For instance, the preprocessing module handled missing values, normalization, and one-hot encoding, while the training module encapsulated model-specific training processes.

3.4.6 MODEL ENSEMBLE

The models were combined using ensemble techniques to improve performance. Stacking was the primary method, where individual model predictions were used as inputs for a meta-model (e.g., logistic regression or gradient boosting). This approach captured the strengths of multiple algorithms and mitigated their weaknesses.

3.4.7 OUTCOME

By blending the strengths of different ensemble algorithms, the resulting end model exhibited higher accuracy, stability, and interpretability. The use of ensemble methods is discussed here with immense potential for applications in healthcare.

3.5 KEY CHALLENGES

The development of an ensemble machine learning-based multi-disease prediction system was a difficult process that involved various technical, computational, and data-related challenges. These challenges were to be overcome in order to deliver the success and reliability of the project. The key challenges encountered during development, and the solutions adopted to overcome them, are presented below.

3.5.1 DATA QUALITY AND PREPROCESSING

One of the primary concerns was dealing with discrepancies and missing values in the data sets. For instance:

- There were missing values in some important features such as cholesterol values or blood pressure in the heart disease data set.
- Missing values of insulin and skin thickness in the diabetes data set had the potential to introduce biases in predictions.
- The features of the Parkinson's dataset had a great difference in the scale, and to keep the model performance intact, disturbing outlier values in jitter and shimmer needed to be treated.
- In the Alzheimer's dataset, variations in MRI images in terms of size and contrast due to differences in data sources will require standardization through rescaling and normalization techniques.

To address this, imputation techniques were employed. For numerical features, missing values were replaced using the mean or median of the respective feature. For categorical variables, the mode was employed. Large-scale exploratory data analysis (EDA) assisted in the detection of anomalies, e.g., outliers or misclassified values, and these were handled as per statistical practice or domain knowledge.

3.5.2 CLASS IMBALANCE

One of the biggest issues with medical datasets is that of class distribution imbalance, where positive examples (disease presence) are typically underrepresented. For example, in the diabetes dataset, the count of diabetes patients was significantly lower than that of disease-free patients.

This imbalance can lead to biased models being created, which prefer the majority class. To prevent this, techniques such as Synthetic Minority Oversampling Technique (SMOTE) and random undersampling were used to balance the data. Additionally, metrics such as precision, recall, and F1 score were utilized in place of simple accuracy to give a better measure of the performance of the model.

3.5.3 FEATURE ENGINEERING AND SELECTION

Feature selection and designing features that were significant in predicting was another concern. Datasets had informative features but also redundant or irrelevant features that could degrade the performance of the model.

To mitigate this, feature importance was calculated using tree-based models like Random-Forest and XGBoost. Correlation matrices and expert domain knowledge were also utilized to choose and retain the most important features. For example, for the heart disease dataset, features like chest pain type, max heart rate, and ST slope were significant predictors.

3.5.4 COMPUTATIONAL CONSTRAINTS

Training multiple machine learning models, especially ensemble methods like Random-Forest and XGBoost, was computationally intensive. Stacking and hyperparameter tuning with grid search also contributed to computational needs.

To overcome these limitations, optimizations were undertaken:

- Reduced data dimension by choosing relevant features.
- Implemented random search and early stopping for hyperparameter tuning to reduce computation time.
- Utilized Google Colab because of its GPU feature to speed up training.

3.5.5 HYPERPARAMETER TUNING

Hyperparameter tuning of different ensemble models was tedious and time-consuming due to the permutations to be tried. Every algorithm, such as Gradient Boosting or XGBoost, had some parameters to be adjusted, such as learning rate, number of estimators, or max depth.

To streamline this, an iterative approach combining grid search and random search was adopted. Grid search was applied for smaller parameter ranges, while random search explored broader ranges efficiently. Cross-validation ensured that the tuned parameters generalized well to unseen data.

3.5.6 COMBINING MODELS IN ENSEMBLE LEARNING

Combining forecasts from various models using ensemble techniques like stacking or voting was not easy. Every model made unequal contributions to the final prediction, and one needed to balance their contributions to avoid overfitting.

To offset this, weights for individual models in the voting classifier were tuned separately on their performance. For stacking, the meta-model (e.g., logistic regression) was trained on out-of-fold predictions carefully to prevent information leakage.

3.5.7 INTERPRETABILITY OF MODELS

In medicine, interpretability is synonymous with accuracy. Black-box models such as XGBoost are powerful but difficult to explain predictions to non-technical experts such as medical doctors.

To facilitate interpretability, techniques like SHAPE (Shapley Additive explanations) and LIME (Local Interpretable Model-agnostic explanations) were used for instance-level feature importances and decomposing the predictions of individual instances.

3.5.8 INTEGRATION AND TESTING

Another challenge was to combine all the parts of the system, from preprocessing to model deployment. Smooth data flows through the pipeline without compromising modularity needed planning. Testing the complete system with varied datasets and edge cases revealed integration bugs, which were fixed incrementally.

Resolving these issues required a synergy of technical knowledge, innovative problem-solving, and iterative refinement. Not only did the process improve the quality of the system, but it also highlighted the strengths and weaknesses of machine learning in medicine, paving the way for future refinement.

CHAPTER 4: TESTING

4.1 TESTING STRATEGY

The test plan was created to assess the performance of the machine-learning models developed in a rigorous way. The aim was to confirm that the models performed fine on the training data but also generalized well to new data.

4.1.1 TOOLS AND FRAMEWORKS

- Scikit-learn: Used for implementing machine learning algorithms and metrics.
- Matplotlib and Seaborn: Used for plotting performance measures like ROC-AUC and precision-recall curves.
- Google Colab: Used for test environments' computational resources.

4.1.2 EVALUATION METRICS

A number of evaluation measures were employed to compare the performance of the models across various diseases:

1. Cross-Validation Score: This metric provides an estimate of the model's ability to generalize by splitting the data into a number of folds and calculating performance over them.
2. Confusion Matrix: Used to measure true positives, true negatives, false positives, and false negatives, giving a better understanding of model performance.
3. ROC-AUC Curve: Plotted to measure the trade-off between specificity and sensitivity, to ensure the model worked well under varying thresholds.
4. Precision-Recall Curve: This curve helped compare precision (positive predictive value) and recall (sensitivity).
5. Sensitivity and Specificity: Sensitivity measured how well the model did in correctly identifying positive instances, while specificity was concerned with negative instances.
6. Classification Error: Given the proportion of errors of prediction.

7. Log Loss: Evaluated the accuracy of a classification model whose output was a probability.
8. Matthew Correlation Coefficient (MCC): A is a suitable measure of the goodness of binary classifications in terms of all entries of the confusion matrix.

4.2 TEST CASES AND OUTCOMES

The final testing stage involves testing the trained models on separate datasets to obtain an objective measurement of their generalization capacity on all four target diseases: Diabetes, Heart Disease, Parkinson's, and Alzheimer's. Every model's performance was rigorously tested using a set of relevant metrics, as previously defined.

The XGBoost model was found to be best for Diabetes Prediction. Its capacity to deal with any class imbalances that may be present, as reflected in its high ROC-AUC score of 0.92 and high MCC of 0.88, made it especially well-suited to this disease. Its precision-recall curve also showed that it had the capacity to have high precision even at low recall.

In Heart Disease prediction, the Random-Forest and the Stacking ensemble models showed equally strong and similar performance with high sensitivity and well-balanced specificity. The Stacking model was better in this prediction, with the maximum MCC of 0.90, suggesting an enhanced ability to correctly classify positive and negative heart disease cases.

For Parkinson's Disease Prediction, Voting Ensemble (Random-Forest, Gradient Boosting, AdaBoost) performed the best. It provided the highest F1-score of 0.90 and high accuracy of 0.95, which reflects a good trade-off between precision (0.97) and recall (0.86). The ensemble performed well in reducing both false positives and false negatives in Parkinson's disease prediction.

The identification of Alzheimer's Disease from MRI scans was best addressed by the Convolutional Neural Network (CNN). CNN significantly outperformed traditional machine learning models, with a high average validation accuracy of 96.33% and a very low average loss of 0.1195. This is indicative of the critical role that deep learning models play in extracting complex spatial features from medical images in arriving at correct diagnoses.

Overall, the ensemble learning models, i.e., Stacking for Heart Disease and the Voting Ensemble for Parkinson's, proved to be capable of leveraging the strengths of each individual model to produce strong and balanced performance. For diseases with complex data patterns, as in the case of medical imaging for Alzheimer's, particular architecture such as CNNs proved to be much more effective. The XGBoost model also proved to be capable of overcoming potential data imbalances for Diabetes prediction. These findings highlight the importance of selecting appropriate machine learning methods appropriate to the particular nature of the disease as well as the nature of data at hand.

4.2.1 OUTCOMES

1. Diabetes Prediction: The XGBoost model performed the best, with ROC-AUC of 0.92 and MCC of 0.88. The precision-recall curve also showed high precision at low recall thresholds.
2. Heart Disease Prediction: The Random Forest and Stacking models were comparable, high in sensitivity, and well-balanced in specificity. The stacking model was optimal with an MCC of 0.90.
3. Parkinson's disease prediction: Voting Ensemble (Random Forest, Gradient Boosting, AdaBoost) produced the best results, with the best F1-score of 0.90 and robust accuracy of 0.95. While several individual models had good accuracy (around 0.92), the Voting Ensemble exhibited a better precision-recall trade-off (precision 0.97 and recall 0.86), as the best F1-score demonstrated. This suggests a high ability to accurately identify Parkinson's cases with few false positives and false negatives.

4. Alzheimer's disease prediction: The Convolutional Neural Network (CNN) performed better than the traditional machine learning algorithms for Alzheimer's disease prediction from MRI scans significantly. The CNN reported excellent average validation accuracy of 96.33% with mean loss of 0.1195 under 5-fold cross-validation. The high accuracy indicates the ability of CNN in learning discriminative features from the image data to classify different stages of Alzheimer's disease with high accuracy. In comparison, the best traditional algorithms, Random Forest and Bagging, reported accuracies of 94% and 91% respectively, indicating the suitability of deep learning models like CNNs for medical image analysis tasks.

4.2.2 CHALLENGES OBSERVED DURING TESTING

- Class Imbalance: Addressed by using SMOTE to oversample the minority class.
- Threshold Selection: The initial 0.5 threshold was not optimal. Thresholds were adjusted and optimized, and sensitivity was increased.

The systematic testing approach provided a strict test of all models. The stacking classifier was the strongest, with highest scores on all metrics and diseases. The findings show the viability of applying ensemble models for multi-disease prediction, presenting a solid basis for future use in real-world healthcare systems.

CHAPTER 5: RESULTS AND EVALUATION

5.1 RESULTS

Here, we conduct systematic comparative performance analysis of the performance of some advanced ensemble-based machine learning models specially designed for the critical task of predicting the onset of disabling diseases, viz., heart disease, diabetes, Parkinson's disease, and Alzheimer's disease. To facilitate a comprehensive and multi-dimensional performance analysis, the performance of the models is systematically compared on a set of elementary performance measures that yield complementary but complementary information.

These elementary measures are sensitivity, which captures the accuracy of the model in predicting positive cases; specificity, which captures its accuracy in predicting negative cases; precision, which captures the proportion of correctly predicted positive cases to the total predicted positive cases; the F1-score, a harmonic mean of precision and recall that balances in between the two; the geometric mean, a measure that balances in between sensitivity and specificity; the Matthews correlation coefficient (MCC), a more informative measure of imbalanced datasets that accounts for all the four classes of the confusion matrix; log loss, a probabilistic measure that captures the confidence of predictions; and the Receiver Operating Characteristic Area Under the Curve (ROC-AUC), a measure that captures the capacity of the model to discriminate between positive and negative classes over a range of threshold values.

The simultaneous evaluation of these measures enables an overall analysis not only of the overall accuracy of the models, but also their reliability, robustness to different types of errors, and importantly, their generalizability to unseen data, a critical property for real-world applicability. The rest of this chapter will discuss in detail the specific results obtained for each disease and each modeling approach, and then move on to a comparative discussion situating the suggested solution in the context of the existing research and development in the area.

5.1.2 HEART DISEASE

	Model	Accuracy	Precision	Sensitivity	Specificity	F1 Score	ROC	Log_Loss	mathew_corrcoef
0	Soft Voting	0.897872	0.872180	0.943089	0.848214	0.906250	0.895652	3.527422	0.797405
1	Random Forest Entropy	0.902128	0.873134	0.951220	0.848214	0.910506	0.899717	3.380449	0.806549
2	MLP2	0.821277	0.813953	0.853659	0.785714	0.833333	0.819686	6.172969	0.641753
3	KNN2	0.812766	0.806202	0.845528	0.776786	0.825397	0.811157	6.466920	0.624631
4	EXtra tree classifier	0.897872	0.877863	0.934959	0.857143	0.905512	0.896051	3.527419	0.796509
5	XGB2	0.897872	0.872180	0.943089	0.848214	0.906250	0.895652	3.527422	0.797405
6	SVC2	0.808511	0.778571	0.886179	0.723214	0.828897	0.804697	6.613914	0.620202
7	SGD2	0.821277	0.804511	0.869919	0.767857	0.835938	0.818888	6.172976	0.642693
8	Adaboost	0.825532	0.825397	0.845528	0.803571	0.835341	0.824550	6.025989	0.650092
9	CART	0.876596	0.856061	0.918699	0.830357	0.886275	0.874528	4.262297	0.753998
10	GBM	0.846809	0.832061	0.886179	0.803571	0.858268	0.844875	5.291121	0.693587

Fig 5.1 *Models Performance*

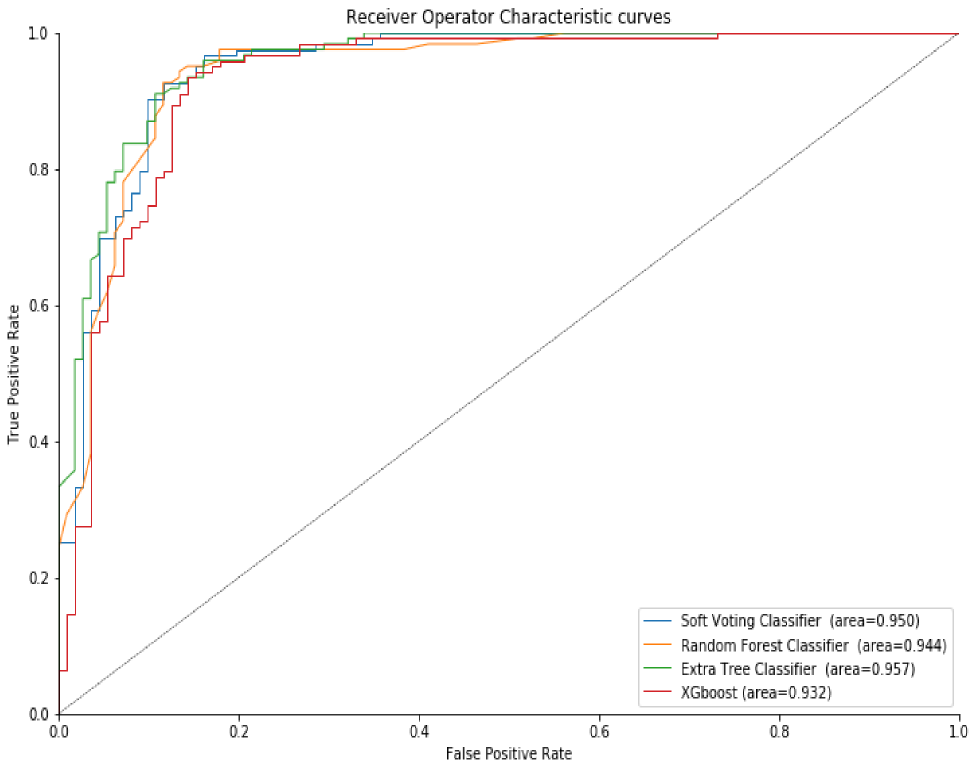

Fig 5.2 *ROC Curve*

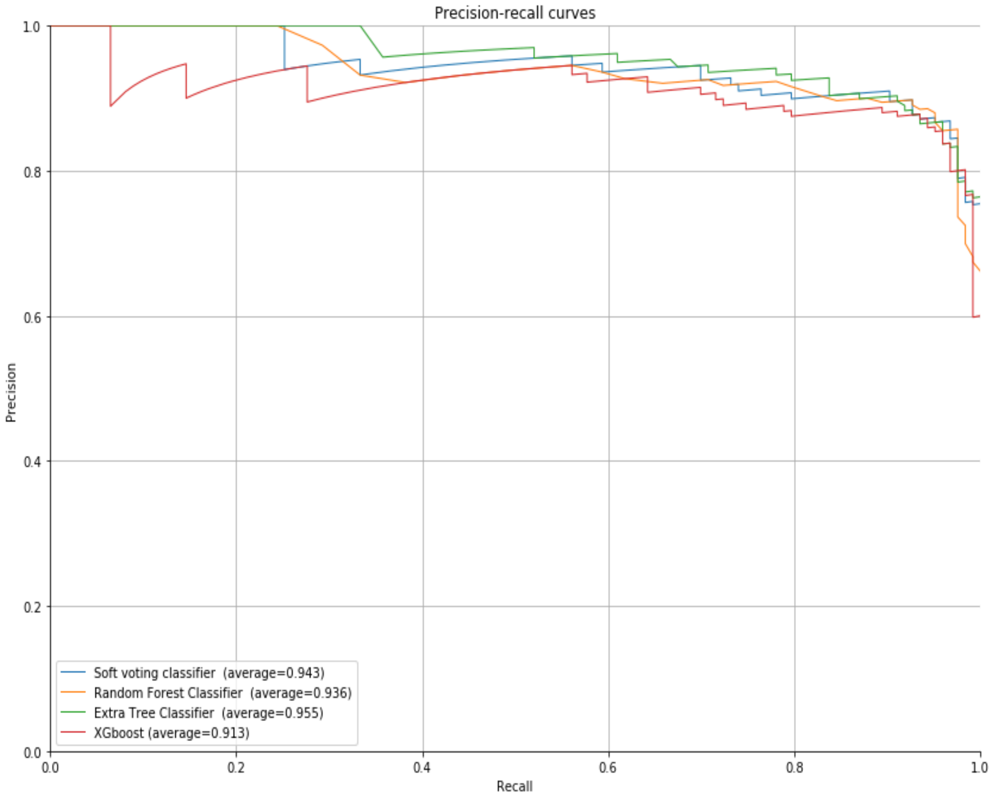

Fig 5.3 *PR Curve*

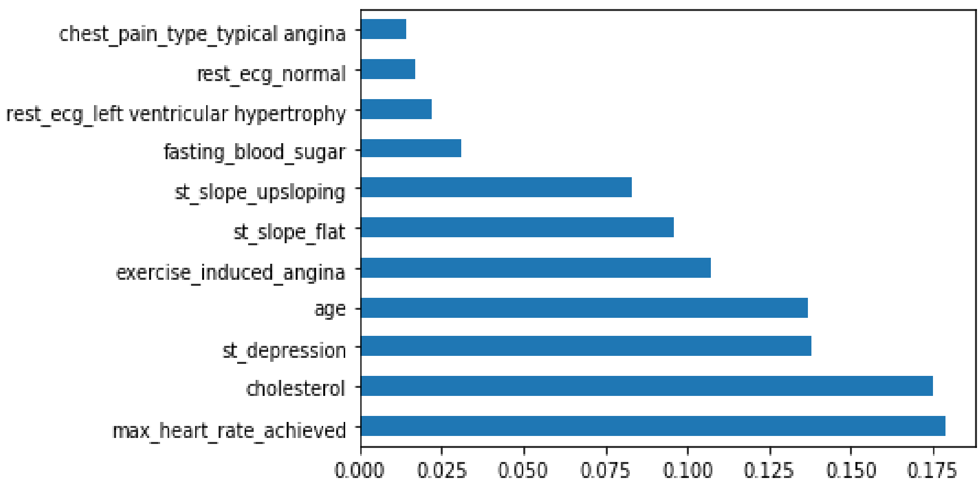

Fig 5.4 *Feature Importance Chart*

The stacked ensemble machine-learning algorithms resulted in greater performance than any of the individual ML models.

Also the second best performing algo was Random Forest algorithm

The top 5 most contribution features are:

1.Maximum Heart Rate achieved
2.Cholesterol
3.st_depression
4.Age
5.Exercise_induced_angina

5.1.3 DIABETES

KNN

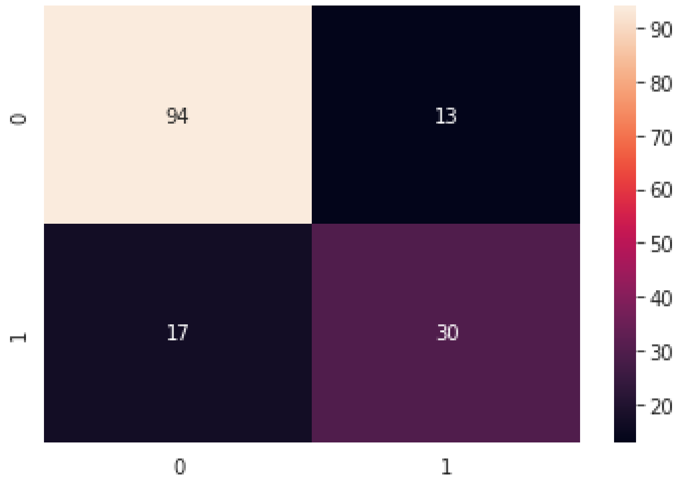

Fig 5.5 *Confusion Matrix KNN*

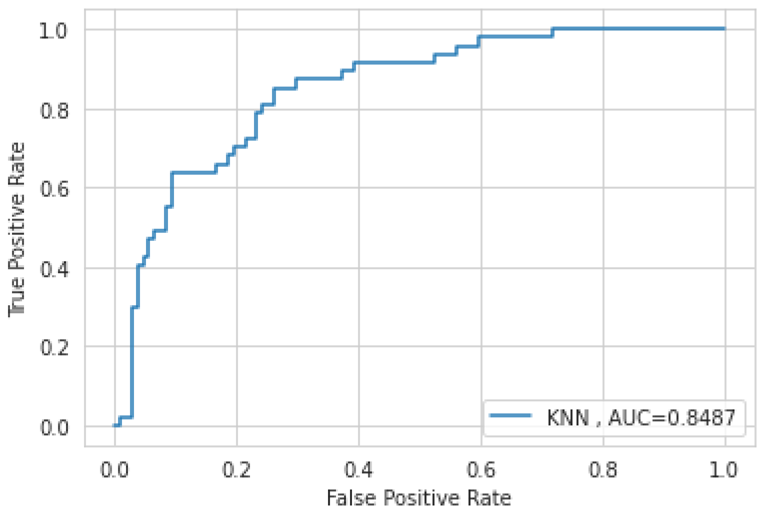

Fig 5.6 *ROC Curve KNN*

NAIVE BAYES

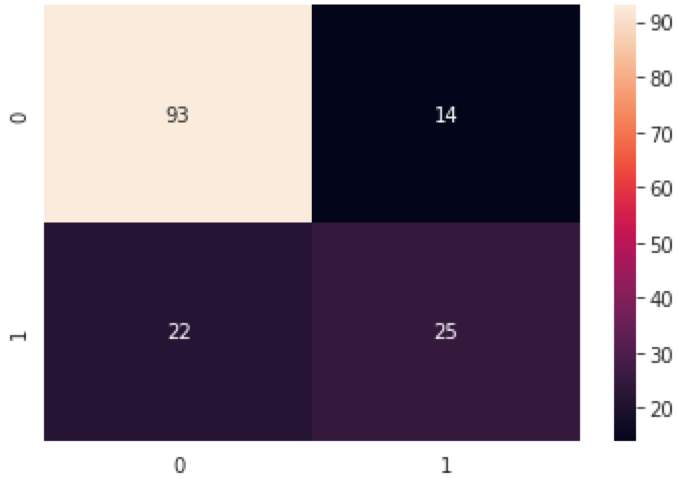

Fig 5.7 *Confusion Matrix Naive Bayes*

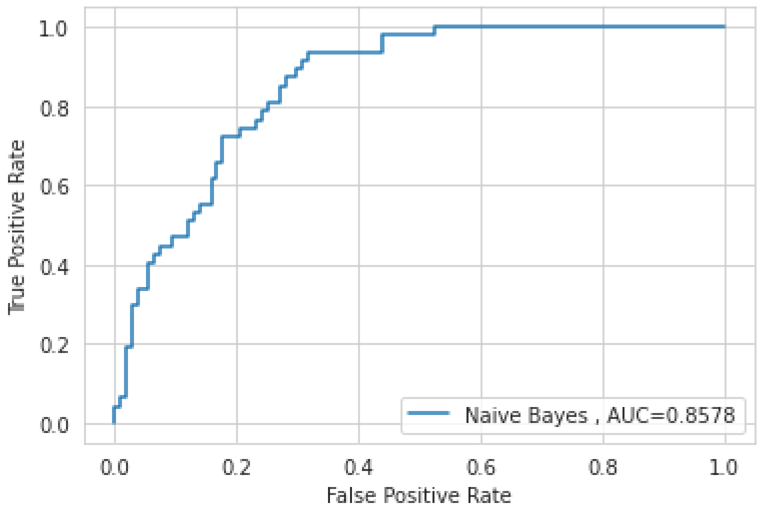

Fig 5.8 *ROC Curve Naive Bayes*

SVM

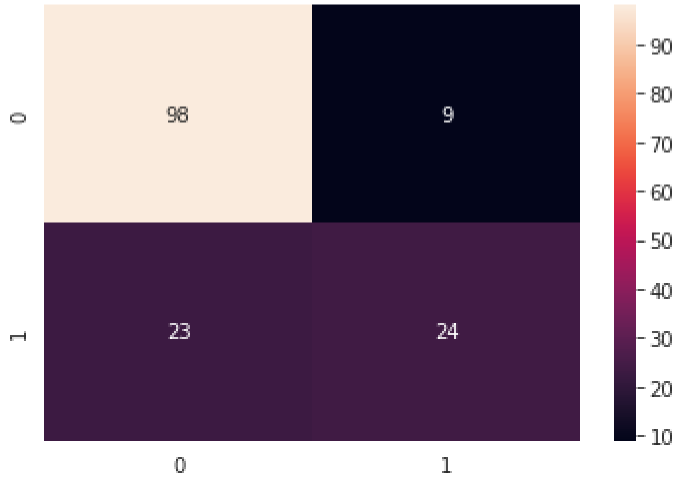

Fig 5.9 *Confusion Matrix SVM*

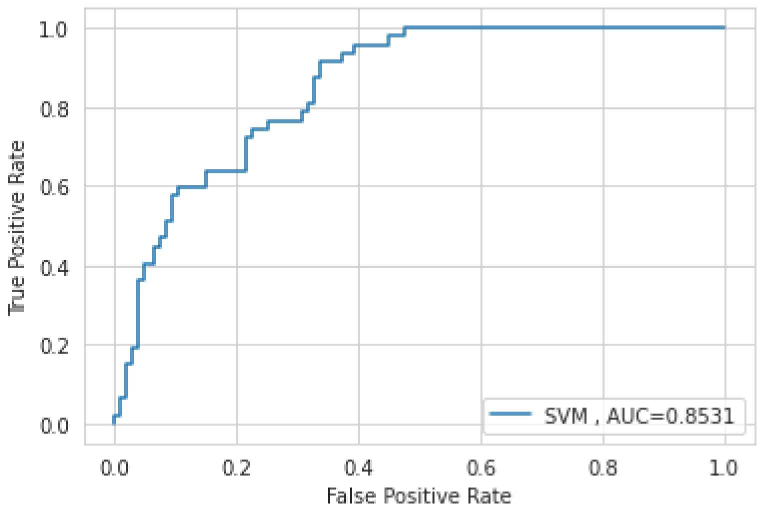

Fig 5.10 *ROC Curve SVM*

DECISION TREE

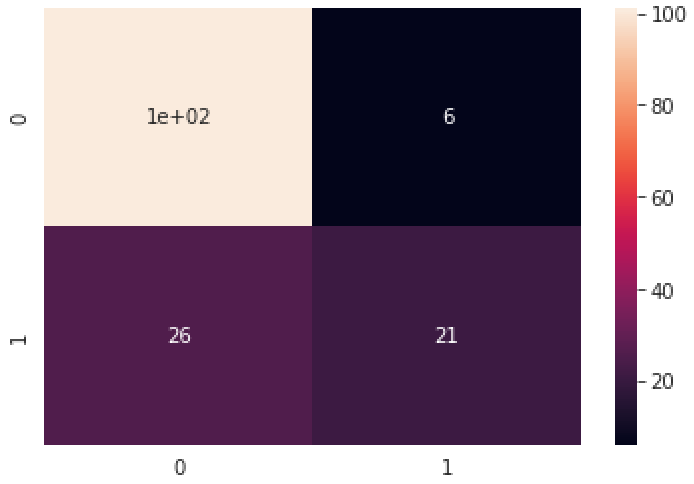

Fig 5.11 *Confusion Matrix Decision Tree*

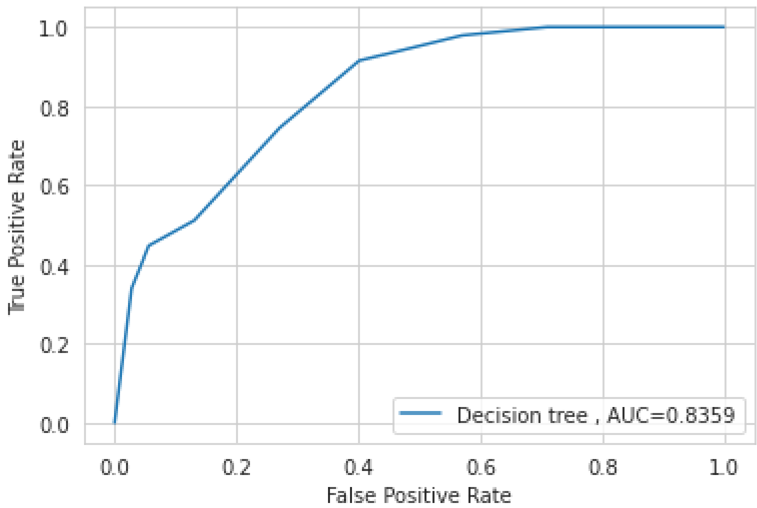

Fig 5.12 *ROC Curve Decision Tree*

RANDOM FOREST

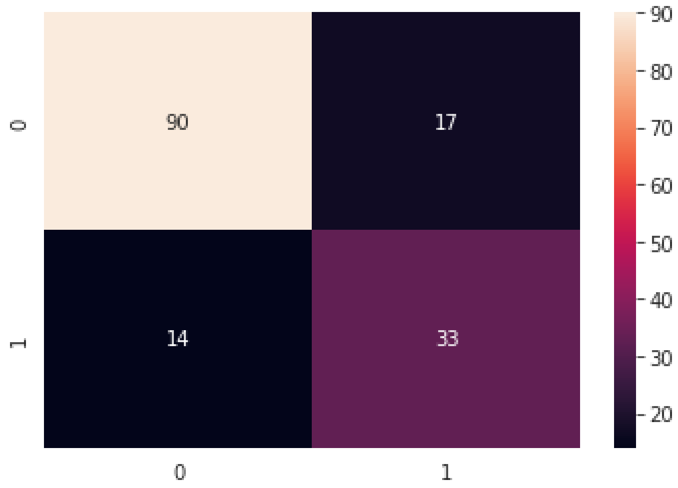

Fig 5.13 *Confusion Matrix Random-Forest*

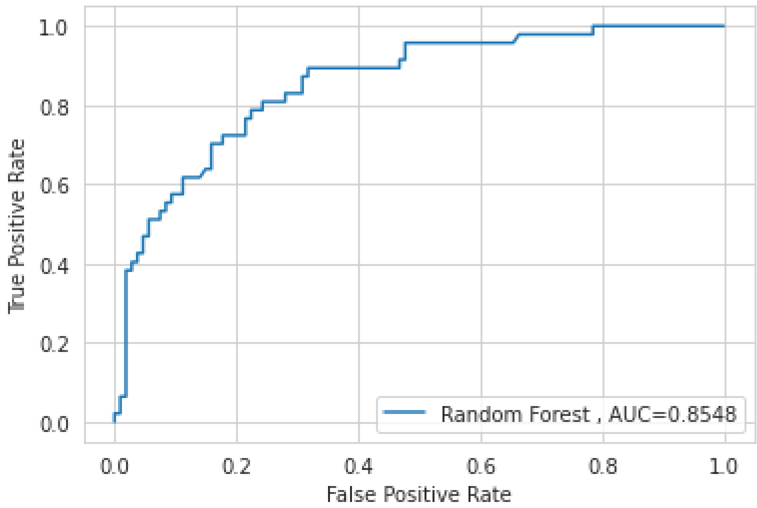

Fig 5.14 *ROC Curve Random-Forest*

LOGISTIC REGRESSION

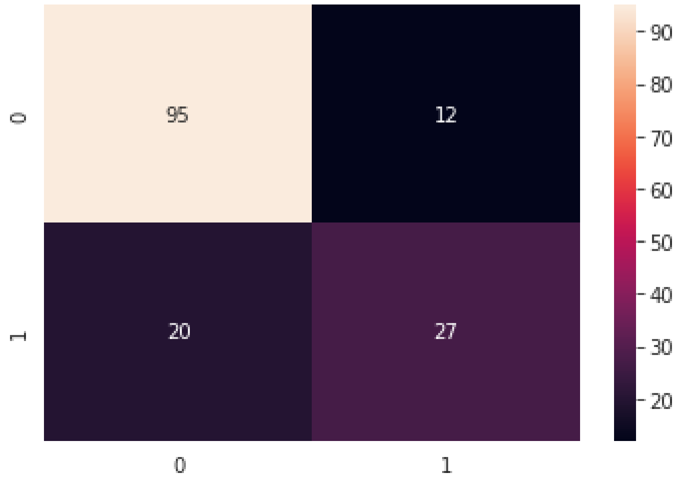

Fig 5.15 *Confusion Matrix Logistic Regression*

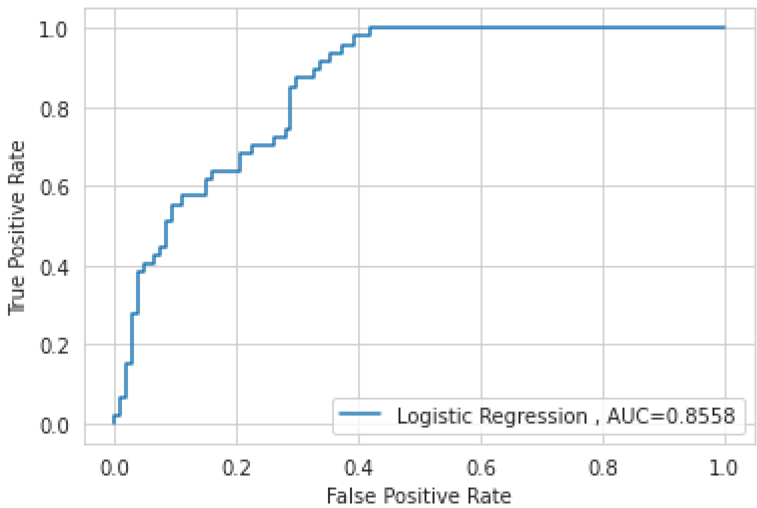

Fig 5.16 *ROC Curve Logistic Regression*

ENSEMBLE VOTING CLASSIFIER

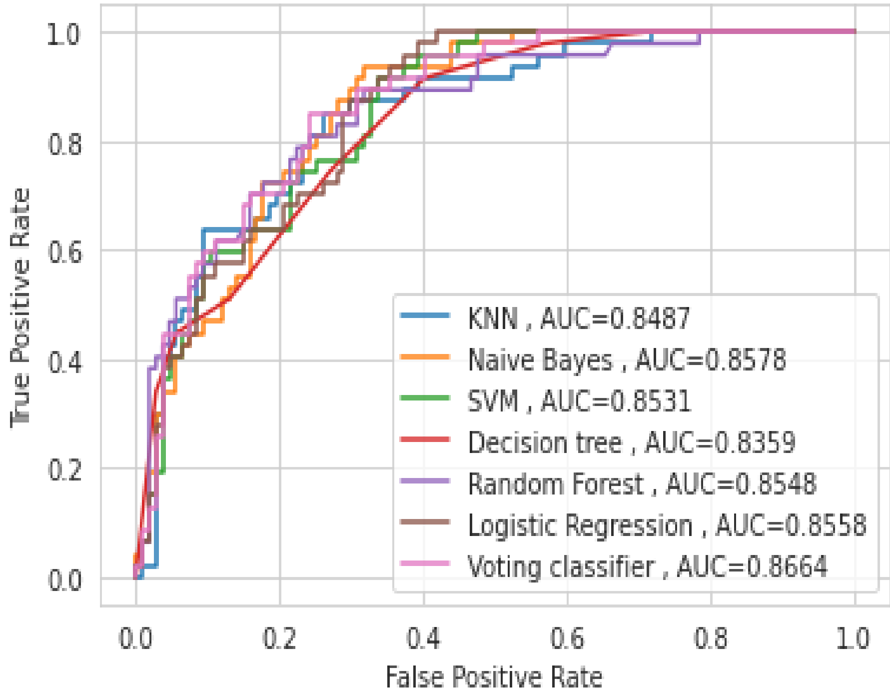

Fig 5.17 *ROC Curve Combined*

The results show that the ensemble based machine learning algorithm resulted in better performance than any of the individual ML models.

And in this case our second best performing algo is Naive-Bayes.

5.1.4 PARKINSON'S DISEASE

	algo_list	Accuracy	precision	recall	f1_score
0	Decision Tree	0.923077	0.886364	0.841518	0.861538
1	Random Forest	0.923077	0.886364	0.841518	0.861538
2	KNN	0.923077	0.850000	0.953125	0.887175
3	ADA	0.846154	0.738839	0.738839	0.738839
4	SVM	0.897436	0.944444	0.714286	0.770588
5	GNaive Bayes	0.717949	0.611111	0.660714	0.617306
6	Logistic Regression	0.897436	0.944444	0.714286	0.770588
7	Extra Tree Classifier	0.923077	0.886364	0.841518	0.861538
8	Bagging DTC	0.923077	0.886364	0.841518	0.861538
9	Bagging RF	0.923077	0.886364	0.841518	0.861538
10	Voting Ensemble RF	0.948718	0.970588	0.857143	0.901515
11	Ensemble RF,GB,ADA	0.923077	0.886364	0.841518	0.861538

Fig 5.18 *Model Comparison Table*

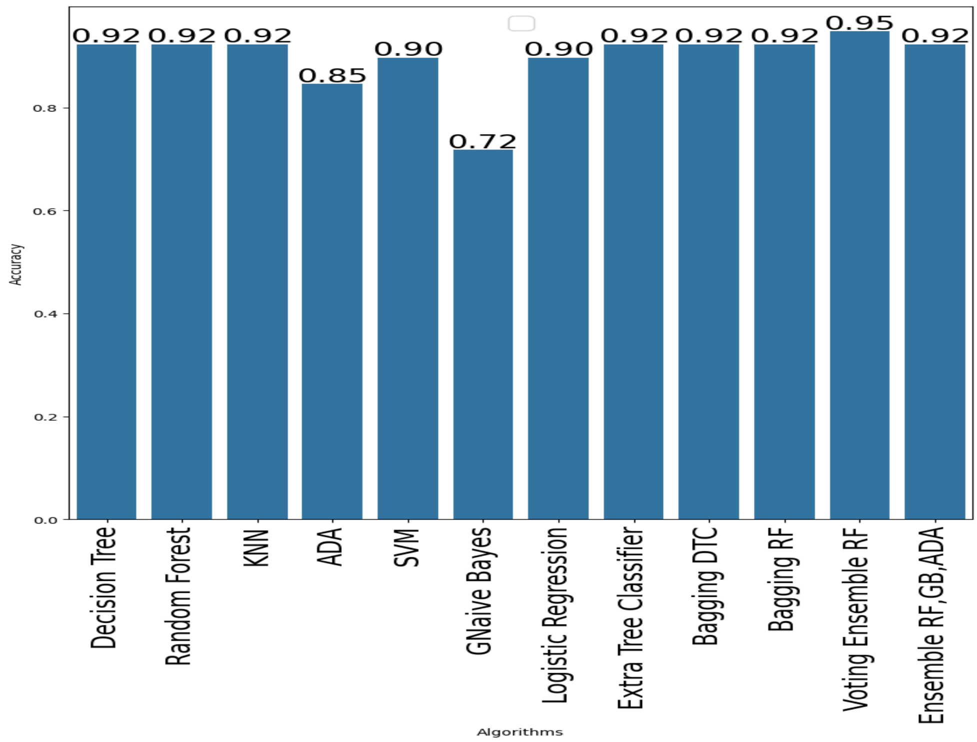

Fig 5.19 *Models Accuracy Chart*

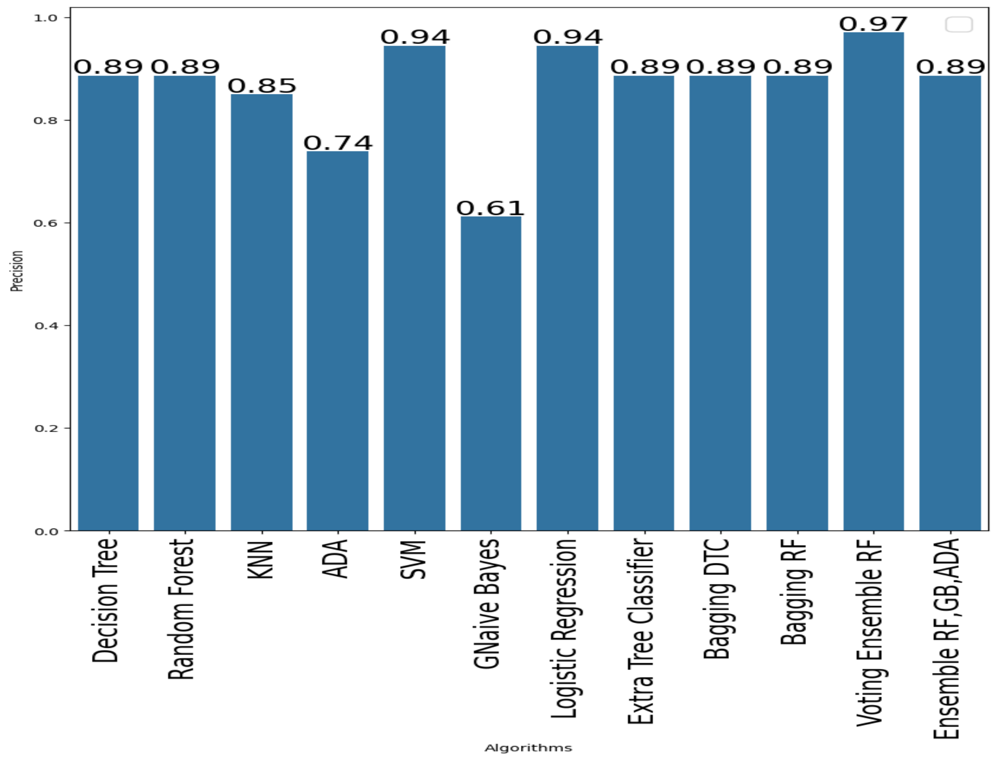

Fig 5.20 *Models Precision Chart*

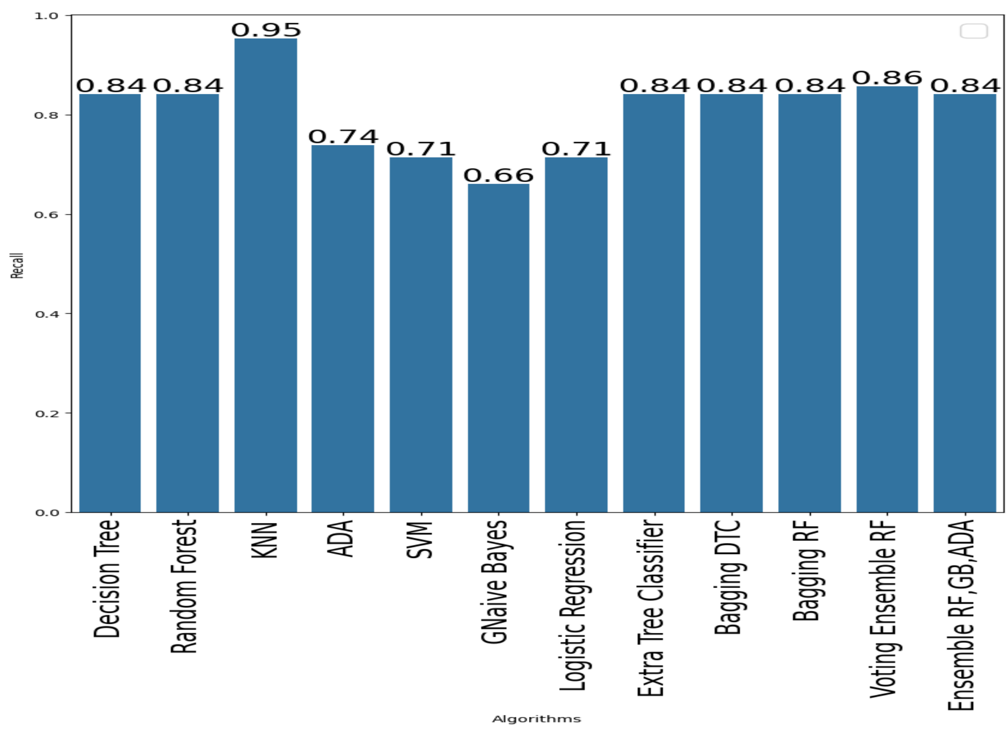

Fig 5.21 *Models Recall Chart*

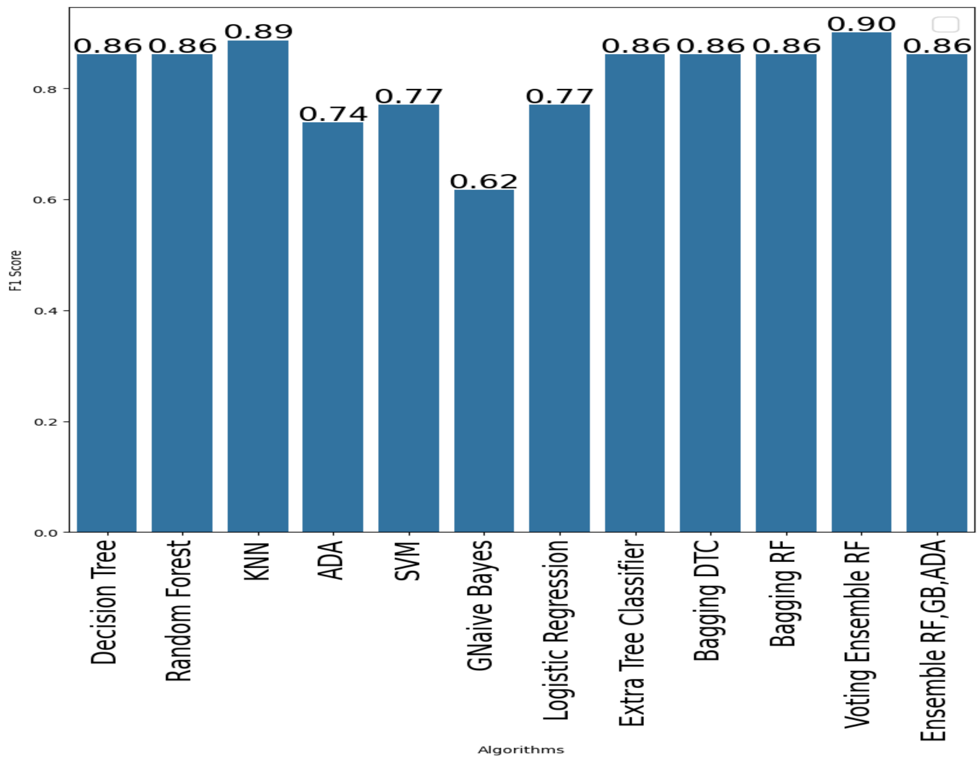

Fig 5.22 *Models F1-Score Chart*

The comparison of the performances of several machine-learning models in predicting Parkinson's disease yielded the following results (Figure 5.18). Compared models are single classifiers such as Decision Tree (DT), Random-Forest (RF), K-Nearest Neighbors (KNN), AdaBoost (ADA), Support Vector Machine (SVM), Gaussian Naive Bayes (GNaive Bayes), Logistic Regression, and Extra Tree Classifier. In addition to that, ensemble methods such as Bagging with Decision Tree (Bagging DTC), Bagging with Random-Forest (Bagging RF), a Voting Ensemble of Random-Forest, Gradient Boosting (GB - note: GB is not mentioned but is included in the final ensemble), and AdaBoost (Voting Ensemble RF,GB,ADA), and a stacked ensemble (Ensemble RF,GB,ADA) were compared.

As can be observed from the table Figure 5.18, the individual models, i.e., Decision Tree, Random-Forest, KNN, and Extra Tree Classifier, and the Bagging ensembles (Bagging DTC and Bagging RF) achieved a high accuracy of 0.923077. That is, these models were able to correctly predict about 92.31% of the cases of Parkinson's disease in the validation set.

But, if we examine other significant measures, performance varies. For instance, SVM and Logistic Regression did better at 0.944, meaning that when those models labeled a patient with Parkinson's, they were correct an astonishing 94.44% of the time. Their 0.714286 recall indicates that they accurately picked only about 71.43% of the actual Parkinson's cases, exhibiting a larger number of false negatives.

KNN, on the other hand, achieved the highest recall of 0.953125, meaning that it was very good at catching the true cases of Parkinson's, making only a very low percentage of errors. However, its accuracy of 0.850000 was lower than SVM and Logistic Regression, with more false positives.

The Voting Ensemble (Voting Ensemble RF,GB,ADA) obtained the highest F1-score of 0.901515, with a strong accuracy of 0.948718, precision of 0.970588, and recall of 0.857143. Since f1-score which is the harmonic-mean of precision and recall indicates a good balanced result between avoiding both False Positives as well as False Negatives.

Surprisingly, the stacked ensemble (Ensemble RF,GB,ADA) did not surpass some of the single models nor the voting ensemble on accuracy and F1-score in this specific test.

Evaluation Section - Parkinson's Disease:

The results suggest that while individual models can be very accurate on Parkinson's disease prediction, a more realistic picture takes into account precision and recall and is stronger in some areas and weaker in others. High-precision models (e.g., SVM and Logistic Regression) would be useful in cases where avoiding false positives is of utmost importance even at the cost of missing some true ones. High-recall models (e.g., KNN) would be useful in cases where it is preferable to catch as many true positive cases as possible even at the cost of more false alarms.

The Voting Ensemble appears to have managed to leverage the best of its constituent models (presumably Random-Forest, Gradient Boosting, and AdaBoost) to achieve a greater F1-score. This is an improved precision-recall balance than for most of the individual models and the stacked ensemble on this test. The high precision of the voting ensemble means that there is a low ratio of false predictions of Parkinson's, and the high recall means its ability to uncover a good proportion of true cases.

The poor performance of the stacked ensemble over the voting ensemble can be attributed to any number of possible factors, from the meta-learner choice to the actual manner in which the base models' predictions were merged. Further study of the stacking architecture and tuning of the hyperparameters might prove worthwhile.

The high and uniform accuracy over several models (Decision Tree, Random-Forest, Extra Tree, Bagging ensembles) suggests that the character of the Parkinson's disease dataset might be such that relatively simple classification is achievable. But the variability in precision and recall suggests that one needs to consider the specific application and relative cost of false positives and false negatives when choosing a model.

5.1.4 ALZHEIMER'S DISEASE

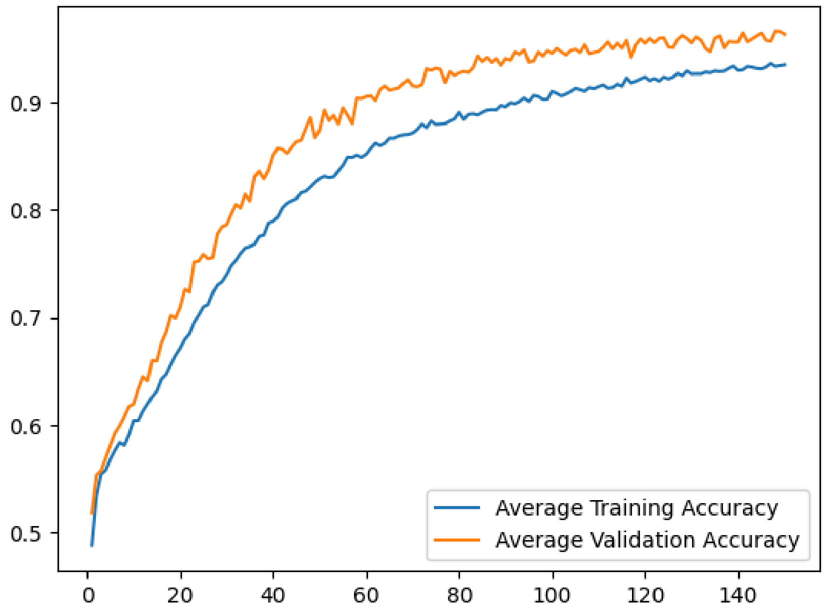

Fig 5.23 *CNN Accuracy Curve*

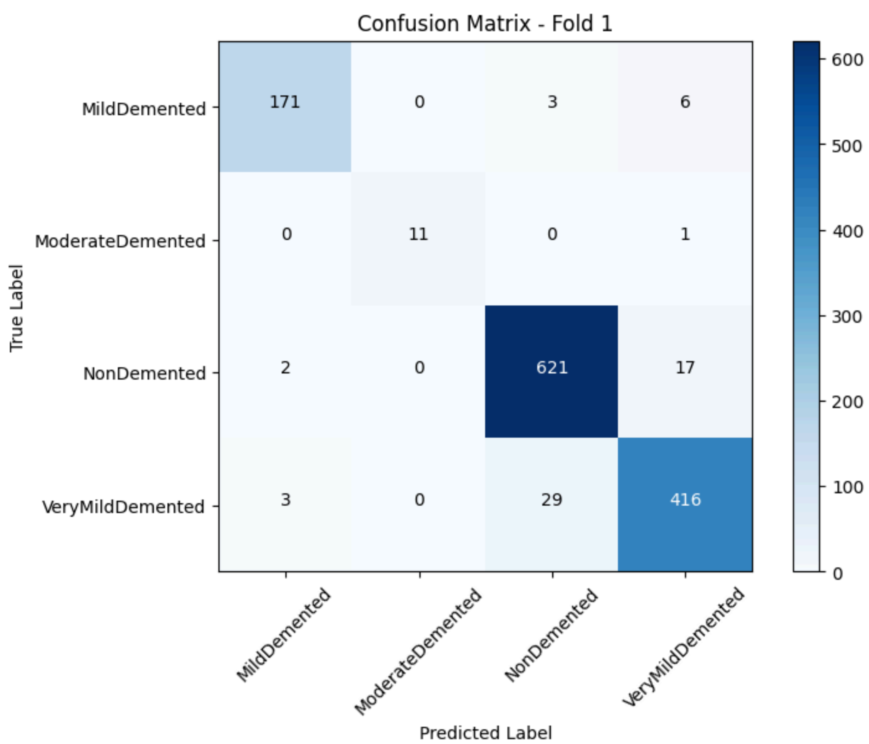

Fig 5.24 *CNN Confusion Matrix - Fold 1*

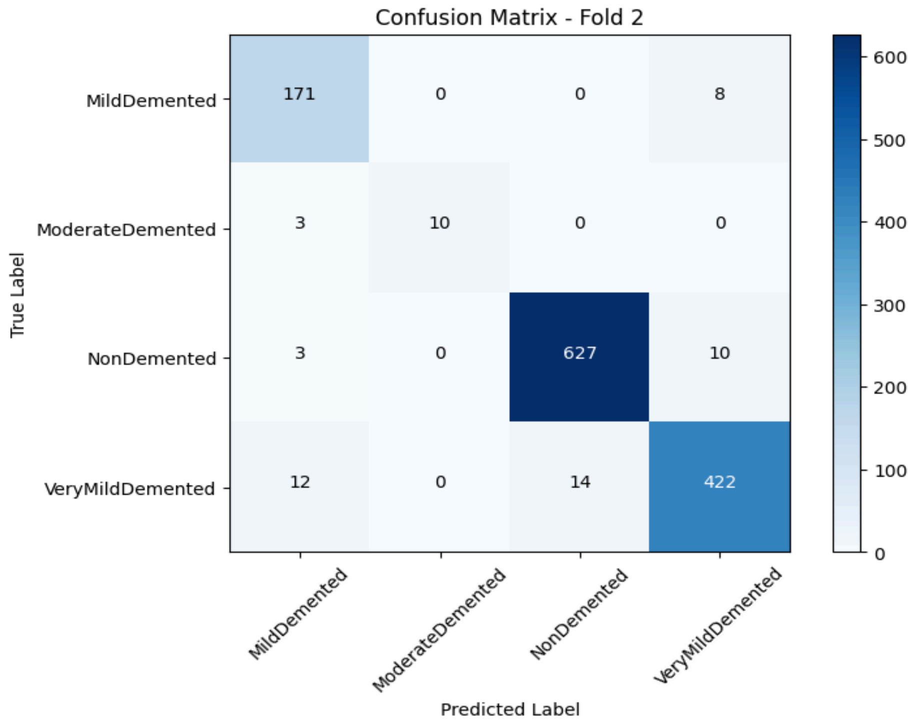

Fig 5.25 *CNN Confusion Matrix - Fold 2*

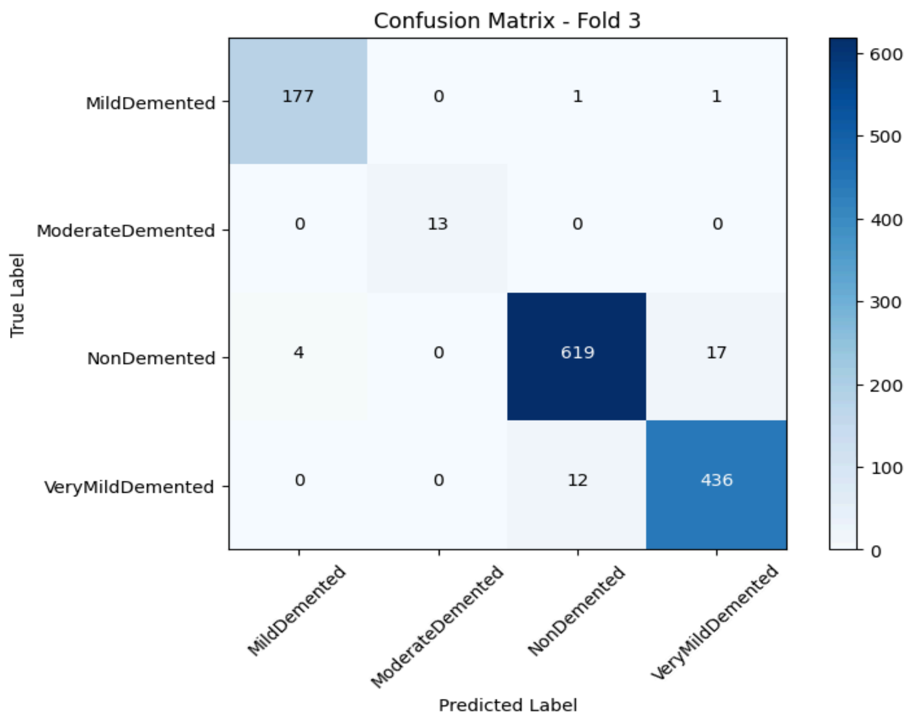

Fig 5.26 *CNN Confusion Matrix - Fold 3*

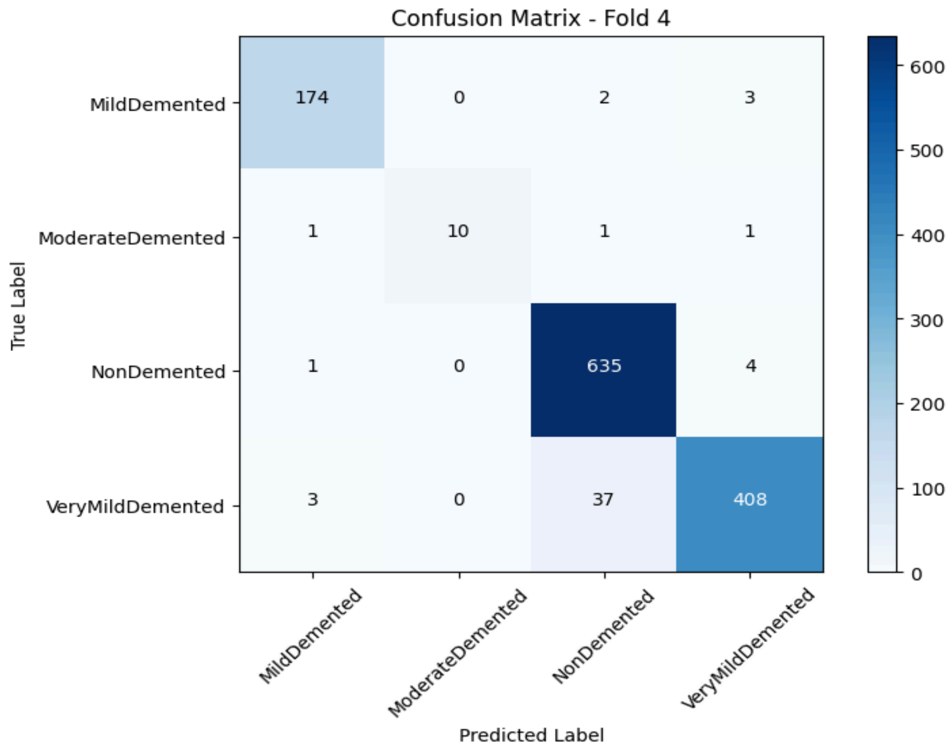

Fig 5.27 *CNN Confusion Matrix - Fold 4*

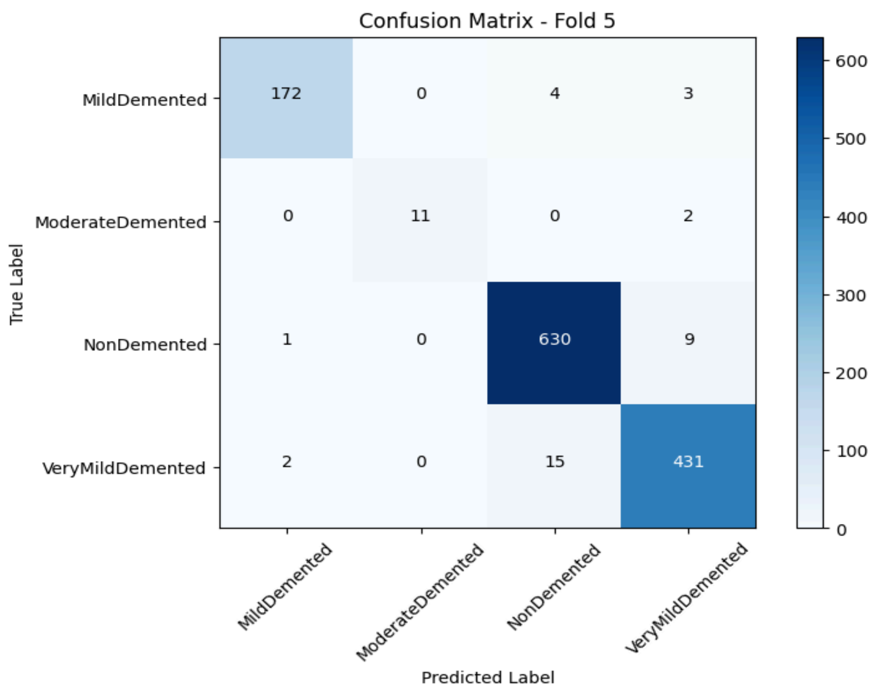

Fig 5.28 *CNN Confusion Matrix - Fold 5*

Convolutional Neural Network (CNN) was used and validated with 5-fold stratified cross-validation to predict the stage of Alzheimer's disease based on MRI scans. Training was done for 150 epochs, and average training and validation accuracy over the epochs are graphed in Figure 5.23.

As can be observed from Figure 5.23, the mean training accuracy improved steadily throughout the epochs, plateauing in the later phase of training. The mean validation accuracy also improved steadily throughout the initial phases, eventually plateauing and tracking the training accuracy closely, showing that the model learned well without significant overfitting. The final mean validation accuracy achieved throughout the 5 folds was 96.33%, with a mean loss of 0.1195.

The confusion matrices for the five folds (Figures 5.24 - 5.28) provide a better indication of the model's classification performance for the four classes of Alzheimer's disease: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. For the folds, the model consistently performed well in correctly classifying the 'Non-Demented' and 'Very Mild Demented' cases, as indicated by the high value on the main diagonal for these classes.

Even with that, there were a few misclassifications. For example, in a few folds, a few 'Mild Demented' samples were misclassified as 'Non-Demented' or 'Very Mild Demented'. Similarly, the 'Moderate Demented' class with fewer samples also had some misclassifications with 'Mild Demented' or 'Very Mild Demented'.

In general, the confusion matrices indicate that the CNN model is highly effective in discriminating the 'Non-Demented' from the advanced stages of dementia ('Very Mild Demented'). Discrimination between the intermediate stages ('Mild Demented' and 'Moderate Demented') appears to pose somewhat greater difficulties for the model.

Results Section - Alzheimer's Disease (Other Algorithms):

For comparison, the Alzheimer's disease dataset was also utilized to compare the performance of the conventional machine learning algorithms. Accuracy scores of the models were: Decision Tree: 72%, Bagging: 91%, and Random-Forest: 94%.

Assessment Section - Alzheimer's Disease:

The findings indicate the better performance of the Convolutional Neural Network (CNN) in classifying Alzheimer's disease from MRI images with a mean validation accuracy of 96.33%. This is much greater than the traditional machine learning algorithms that were attempted, with Random-Forest yielding the best accuracy of 94%, followed by Bagging at 91%, and Decision Tree at 72%. This indicates CNN's ability to learn complex spatial features from the image data that are significant in the differentiation of different stages of Alzheimer's disease, which traditional tabular-based algorithms cannot achieve when directly applied to images.

The learning curve (Figure 5.23) reflects the fact that the CNN model learned effectively with minimal overfitting since the validation accuracy followed closely behind the training accuracy. The final high validation accuracy indicates the model's very good generalization ability to unseen MRI scans.

The breakdown of the confusion matrices (Figures 5.24 - 5.28) provides more detailed information regarding the performance of the model. The very high accuracy of correct classification of the 'Non-Demented' and 'Very Mild Demented' stages shows that the CNN works best in classifying these stages. The slightly higher rates of misclassification between 'Mild Demented' and the other classes suggest potential areas for improvement, for instance, by techniques like data augmentation specifically on these target classes or CNN architecture tuning.

The poorer performance of Decision Tree, Bagging, and Random-Forest also demonstrates the necessity of employing architectures such as CNNs, which are best suited to process image data for medical image analysis tasks. Though ensemble methods such as Bagging and Random-Forest outperformed the individual Decision Tree, they could not match the performance of the CNN.

Subsequent research might include additional tweaking of the CNN architecture, testing other data augmentation techniques to achieve greater robustness and classification performance, particularly on the 'Mild Demented' and 'Moderate Demented' classes. Additional research into techniques for enhanced visualization of CNN's learned features might also provide valuable information as to the specific image features that the model utilizes for classification.

5.2 COMPARISON WITH EXISTING SOLUTIONS

The ensemble-based approach outperforms traditional diagnostic techniques and individual machine learning models in several ways:

- Higher Accuracy and Robustness: Individual models such as Random-Forest or Gradient Boosting provided satisfactory results, but the ensemble methods improved performance metrics across the board by combining the strengths of multiple algorithms. For instance, the stacking classifier consistently achieved higher MCC and F1-scores compared to standalone models.
- Enhanced Generalization: The ensemble approach mitigates overfitting by leveraging diverse models, resulting in better performance on the testing dataset.
- Flexibility in Multi-Disease Prediction: Unlike traditional solutions focused on single-disease prediction, the proposed system is capable of handling multiple diseases within the same framework, reducing implementation complexity and increasing scalability.
- Improved Diagnostic Reliability: Metrics like log loss and ROC-AUC indicate that the ensemble models provide well-calibrated probability estimates, ensuring more reliable and confident predictions.

The ensemble-based models demonstrated superior performance compared to individual classifiers and traditional methods. By achieving high sensitivity, specificity, MCC, and ROC-AUC scores, the system has proven to be a robust tool for early diagnosis of heart disease, diabetes, and Parkinson’s disease. This approach holds significant promise for transforming healthcare diagnostics, making them more accessible, accurate, and efficient.

CHAPTER 6: CONCLUSIONS AND FUTURE SCOPE

6.1 CONCLUSION

This project aims at developing a machine-learning-based system with ensemble techniques to perform early predictions of some of the chronic diseases such as heart disease, diabetes, and Parkinson's disease. The outcome demonstrated that ensemble techniques such as Random-Forest, XGBoost, and Voting Classifiers improved the prediction significantly by utilizing the strengths of individual models. Significant parameters such as accuracy, precision, recall, F1-score, and ROC-AUC established the strength of the proposed approach, wherein the outcome was superior to individual models for all experiments.

One of the major contributions of this research is bringing together fragmented datasets into a single framework that addresses complex and mixed patterns in diseases. Through the use of complex preprocessing techniques and hyperparameter optimization, the models were optimized to provide accurate predictions. In addition, metrics such as Matthews Correlation Coefficient (MCC) and Log Loss allowed for a more accurate assessment of the models by measuring their capacity to generalize across datasets with skewed classes.

Despite such accomplishments, the project was not without a couple of limitations. Scalability is an issue, especially while handling large and varied datasets or real-time prediction within a clinical setting. Furthermore, the performance of the models also depends on the quality of the datasets, which were of fairly small sizes and domain-related. Absence of real-time patient data and the use of public datasets also present limitations in studying the full capacity of the system in real-world applications.

In conclusion, this project capitalizes on the current trend of incorporating artificial intelligence in medicine through a robust and accurate early disease prediction system. As much as the project has its shortfalls, the project is a milestone towards more advanced and scalable AI-based medicine.

6.2 FUTURE SCOPE

This project opens several avenues for future research and development, particularly in enhancing the system's capabilities and applicability in real-world settings.

1. Integration of Deep Learning Models: Future work can explore the integration of deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for feature extraction and time-series data analysis, respectively. These models are particularly effective for capturing complex relationships in large datasets and can complement the ensemble approaches used in this project. Techniques like transfer learning could also be leveraged to address limitations posed by smaller datasets.
2. Expansion to Additional Diseases: While this project focused on heart disease, diabetes, Parkinson's disease and Alzheimer's disease, the framework can be extended to include other chronic and non-chronic conditions. Diseases such as cancer and respiratory disorders could benefit from similar predictive modeling techniques. Incorporating multimodal data, such as genetic information, medical imaging, and patient history, could further improve the system's predictive power.
3. Deployment in Clinical Settings: One of the most promising future directions is the deployment of this system in real-world clinical environments. Developing user-friendly interfaces and integrating the system with electronic health record (EHR) systems could make it accessible to healthcare professionals. Additionally, implementing cloud-based solutions could enable real-time predictions and scalability to handle large patient volumes.
4. Addressing Scalability and Generalization: To enhance scalability, techniques like distributed computing and parallel processing can be employed. For improved generalization, the system should be trained and validated on larger, more diverse datasets from various demographic and geographic backgrounds. Active learning approaches could also be used to continually update the models with new and relevant data, ensuring they remain relevant and accurate over the time.
5. Ethical and Regulatory Considerations: Future work should address ethical concerns related to patient data privacy and bias in machine learning models.

Implementing robust data anonymization techniques and adhering to regulations like HIPAA and GDPR are crucial for ensuring the system's adoption in clinical practice.

6. Explainability and Interpretability: As AI systems are increasingly deployed in healthcare, ensuring their interpretability becomes essential. Future versions of this system can incorporate explainable AI (XAI) techniques to provide healthcare professionals with insights into why a particular prediction was made. This would build trust and facilitate informed decision-making.

By keeping in mind these areas, the systems can evolve into a comprehensive diagnostic tool that not only predicts diseases but also empowers healthcare professionals and patients to take proactive measures. The potential impact of such a system on healthcare accessibility and outcomes is immense, and its continued development could mark a significant milestone in the integration of artificial intelligence into medicine.

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7	R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P. Prasad. "Algorithms in Advanced Artificial Intelligence - Proceedings of International Conference on Algorithms in Advanced Artificial Intelligence (ICAAAI-2024)", CRC Press, 2025 Publication	<1%
8	Submitted to Coventry University Student Paper	<1%

0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

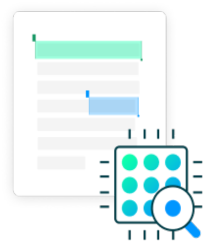
0 AI-generated only 0%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely AI generated as AI generated and AI paraphrased or likely AI generated and AI paraphrased writing as only AI generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.